



OIL INDUSTRY SAFETY DIRECTORATE

**पेट्रोलियम डिपो, टर्मिनलों, कवर्नर्स, सीटीएफ, पाईप लाइन प्रतिष्ठानों
और ल्यूब ऑइल प्रतिष्ठानों की अग्नि संरक्षण सुविधाएं
ओ आई एस डी-मानक -117**

**FIRE PROTECTION FACILITIES
FOR PETROLEUM DEPOTS, TERMINALS, CAVERNS, CTF
PIPELINE INSTALLATIONS AND LUBE OIL
INSTALLATIONS
OISD-STANDARD-117**

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PREFACE

Indian petroleum industry is the energy lifeline of the nation, and its continuous performance is essential for prosperity of the country. As the industry deals with inherently inflammable substances throughout its value chain – upstream, midstream, and downstream – Safety is of paramount importance to this industry as safe performance at all times can ensure optimum return of these national assets and resources including sustainability.

To ensure proper implementation of the various aspects of safety in the oil & gas industry, the Government of India, vide Resolution No. R-13013/4/84-OR-1 dated 10th January 1986 set up a Safety Council at the apex, under the administrative control of the Ministry of Petroleum & Natural Gas (MoP&NG). To assist the Safety Council, a technical directorate, namely the Oil Industry Safety Directorate (OISD) under the aegis of MoP&NG, has been entrusted with the responsibility of formulating standards, overseeing its implementation through safety audits to enhance safety level in the oil & gas industry.

OISD has developed a rigorous, multi-layer, iterative and participative approach for development of standards. OISD develops and revises Standards, Guidelines & Recommended Practices for the oil and gas sector through a participative process involving all the stakeholders, drawing inputs from national and international standards, and adapting them by leveraging the experience of the functional committee. These standards cover inbuilt design safety, asset integrity and best operating practices in the field of production, processing, storage, and transport of petroleum.

The participative process followed in standard formulation has resulted in excellent level of compliance by the industry. It also goes to prove the old adage that self-regulation is the best regulation. The quality and relevance of OISD standards had been further endorsed by their adoption/ reference in various statutory rules and regulations of the land.

OISD and industry together strive to achieve nil incident in the entire hydrocarbon value chain. We at OISD are confident that the provisions of this standard, when implemented in totality, would go a long way in ensuring safe operation in the oil and gas industry.

**Executive Director
Oil Industry Safety Directorate**

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Wherever Acts/ Rules/ Regulation and National/ International Standards are mentioned in the standard, the same relates to in-vogue version of such documents. Whenever there is any difference between the statutory norm and OISD standard, more stringent requirement should be followed. However, in case of any difference in the requirement between OISD standards, the requirement as specified in the standard released at a later stage shall prevail.

The industry has to carry out gap analysis and make necessary changes in their system in line with the revised standard and the requirement should be implemented within a time frame of 1 year (from the date of release of revised standard).

Though every effort has been made to assure the accuracy and reliability of the information contained in these documents, OISD hereby expressly disclaims any liability or responsibility for loss or damage resulting from the use of OISD standard.

The standards would be reviewed periodically based on the decision in the Steering Committee/ Safety Council, industry request, technological development, recommendations of high-level committee and accident investigation reports. Generally, OISD standards are reviewed and revised, reaffirmed, or withdrawn at least every ten years.

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OISD recognizes the support & cooperation of several other experts from industry, who contributed to the preparation, review and finalization of this document.

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1.0 INTRODUCTION

The petroleum depots, terminals & pipeline/lube oil installations are critical in oil industry supply chain to ensure effective supply and distribution. A risk assessment for each installation shall be conducted at design stage to determine the requirement of detection and alarm annunciation of hydrocarbon in atmosphere indicative of unintended release, fire / smoke in enclosed spaces and fire-fighting equipment to safeguard the installations from any fire hazard. Notwithstanding the requirement arising from above risk assessment, minimum safety devices and fire-protection facilities as indicated in the following sections for different type of installations shall be implemented and maintained.

2.0 SCOPE

- 2.1** This standard lays down the minimum requirement of fire protection facilities at Petroleum Depots, Terminals, Caverns, Pipeline Installations with or without Storages, Cross country natural gas pipeline stations, sectionalizing valve stations, Compressor stations, Central Tank Farms (CTF), Lube Oil Installations, Grease Manufacturing & Filling Facilities. The fire protection and firefighting requirement for the combined POL and LPG facilities in the same premises.
- 2.2** This standard shall override the requirements of fire protection facilities specified in any other OISD standard for the installations mentioned in Clause 2.1.
- 2.3** This standard does not cover the fire protection facilities for: -
- i) Depots, Terminals/Installations inside the Refineries and/or Oil/Gas Processing Plants under the same management for which OISD-STD-116 shall be referred.
 - ii) Installations handling Liquefied Petroleum Gas (LPG) Storage, Handling and Bottling. The same are covered in OISD-STD-144, OISD-STD-150 or OISD-STD-169 as the case may be.
 - iii) Drilling rigs, Work over rigs and Production installations which are covered in OISD-STD-189.
 - iv) Port Terminals for which OISD-STD-156 shall be referred. For AFS terminals OISD-STD-135 shall be referred.
 - v) Installations handling LNG and refrigerated LPG.

3.0 DEFINITIONS

3.1 CAVERN

A large cavity either existing or created underground in the side of a hill for storage of crude & LPG.

3.2 CENTRAL TANK FARM /CRUDE OIL TANK FARM(CTF/COT)

It is a facility where petroleum is received and stored in bulk for further dispatch either by

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pipeline, rail, or road. The aggregate storage capacity of petroleum should be above 5000 KL for class A and B petroleum products.

3.3 CLASSIFICATION OF PETROLEUM PRODUCTS

a) General Classification of Petroleum Products:

Petroleum products are classified according to their closed cup FLASH POINTS as follows:

Class 'A' Petroleum: Liquids having flash point below 23°C.

Class 'B' petroleum: Liquids having flash point of 23°C and above but below 65°C.

Class 'C' petroleum: Liquids having flash point of 65°C and above but below 93°C.

Excluded Petroleum: Liquids having flash point of 93°C and above.

Liquefied gases, including LPG do not fall under this classification but form a separate category.

b) Classification for heated products : At locations where the handling temperatures are higher than the flash point of the product in circumstances where product handled is artificially heated above its flash point, excluded product shall be considered as Class 'C' product, Class 'C' product shall be considered as Class 'B' product and Class 'B' product shall be considered as Class 'A' product.

3.4 CLASSIFICATION OF FIRE

Class A Fire: Involve combustible material, such as wood, paper, rubber, and many plastics, etc., where the cooling effect of water is essential for extinction of fire.

Class B Fire: It involves flammable liquids, petroleum products, or the like, where a blanketing effect is essential.

Class C Fire: Involve flammable gases under pressure including liquefied gases where it is necessary to inhibit the burning gas at a fast rate with an inert gas, powder, or vaporizing liquid for extinguishment.

Class D Fire: involve combustible metals such as magnesium, aluminium, zinc, sodium, potassium. The burning metals are reactive to water and water containing agents, and in certain cases carbon dioxide, halogenated hydrocarbons, and ordinary dry powders. These fires require special media and technique to extinguish.

Class F Fire: involve cooking oils and fats, or the like, where a wet chemical fire extinguisher is essential.

3.5 CLEAN AGENT

Electrically non-conductive, volatile, or gaseous fire extinguishants that do not leave a residue upon evaporation and meets the requirements given in the latest NFPA Standard 2001 (latest edition) on clean agent fire extinguishing system in line with environmental consideration of Kyoto & Montreal protocols and latest MoEFCC guidelines.

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3.6 EMERGENCY

Emergency means a situation or scenario which has the potential to cause serious danger to person, the environment, or damage to property and which tends to cause disruption inside or outside the premises and may require the help of outside resources.

CLASSIFICATION OF EMERGENCIES: -

Emergencies can be categorized into three broad levels based on seriousness and response requirements, namely: -

(a) Level - I: This is an emergency or an incident which

(i) can be effectively and safely managed, and contained within the site, location, or installation by the available resources.

(ii) has no impact outside the site, location, or installation.

(b) Level - II: This is an emergency or an incident which

(i) cannot be effectively and safely managed or contained at the location or installation by available resource and additional support is alerted or required.

(ii) is having or has the potential to have an effect beyond the site, location, or installation and where external support of mutual aid partner may be involved.

(iii) is likely to be a danger to life, the environment, industrial assets, or reputation.

(c) Level - III: This is an emergency or an incident with off-site impact which could be catastrophic and is likely to affect the population, property, and environment inside and outside the installation, and management and control is done by district administration. Although the Level III emergency falls under the purview of District Authority but till they step in, it should be the responsibility of the unit to manage the emergency.

Note: Level - I and Level - II shall normally be grouped as Onsite emergency and Level - III as an Off-site emergency.

3.7 FLASH POINT

The minimum temperature at which the liquid gives sufficient vapour, when mixed with air, forms an ignitable mixture, and gives a momentary flash on application of a small pilot flame under specified test conditions.

3.8 HAZARDOUS AREA

An area will be deemed to be hazardous where: -

Petroleum having flash point below 65°C or any flammable gas or vapour in a concentration capable of ignition is likely to be present.

Petroleum or any flammable liquid having flash point above 65°C is likely to be refined, blended or stored at above its flash point.

For classification and extent of hazardous area, refer "The Petroleum Rules - 2002".

3.9 HVLRM

High-volume long-range monitor having capacity 1000GPM(228 m³/hr) and above.

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3.10 INFRASTRUCTURE / OTHER FACILITIES

These are the facilities such as Control Room Building, Sub-Station, Diesel Generator (with diesel storage tank), & Administrative Building, etc. provided in Petroleum Depots, Terminals, Pipeline Installations.

3.11 LUBE OIL INSTALLATIONS

The facilities meant for receipt, storage and blending of base oils & additives into finished Lube products. It includes lube-blending plants, grease manufacturing plants & small can filling plants.

3.12 PETROLEUM

Means any liquid hydrocarbon or mixture of hydrocarbons and any inflammable mixture (liquid, viscous or solid) containing any liquid hydrocarbon.

3.13 PETROLEUM DEPOTS & TERMINALS

A portion of the property, where combustible/flammable liquids are received by tanker, iso containers, pipelines, tank wagons, tank trucks and are stored or blended in bulk for the purpose of distribution by tankers, pipelines, tank wagons, tank trucks, portable tanks or containers.

3.14 PIPELINE INSTALLATIONS

Pipeline Installations are those facilities on cross-country pipelines which have pumping and/or delivery station with or without storages.

3.15 TANK

‘Tank’ means a receptacle for petroleum exceeding 1,000 liters in capacity.

3.16 GENERAL TERMINOLOGY

Shall indicate that provision is mandatory.

Should indicate that provision is recommendatory as per good engineering practices.

MAY: ‘May’ indicates desirable or good practices which are encouraged to be implemented but not necessary.

GPM denotes US gallons (1GPM=3.785 LPM)

4.0 FIRE PROTECTION PHILOSOPHY

The fire protection philosophy is based on loss prevention & control. It considers that the installations as covered in scope carry an inherent potential hazard due to flammable nature of petroleum products stored therein.

Fire shall be controlled/extinguished as quickly as possible to minimize the loss of life, property & damage to environment .

4.1 GENERAL CONSIDERATIONS

The size of facility, pressure and temperature conditions, size of storage, plant location and

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terrain determine the basic fire protection need.

Fire protection of building shall comply to Statutory regulations, as applicable.

4.1.1 LAYOUT

Layout of a depot or terminal, pipeline installation, CTF, lube oil installation, grease manufacturing & filling facilities and handling/disposal system of blow down, drain from equipment handling flammable liquids shall be done in accordance with OISD-STD-118 and other OISD standards as applicable.

4.1.2 FIRE PROTECTION

The following fire protection facilities shall be provided depending upon the nature of the installation and risk involved.

- Fire Water System.
- Foam System.
- Clean Agent Protection System.
- First Aid Fire Fighting Equipment.
- Mobile Fire Fighting Equipment.
- Carbon Dioxide System
- Dry Chemical Extinguishing System
- Portable firefighting equipment
- Detection and Alarm System
- Communication System

4.2 DESIGN CRITERIA FOR FIRE PROTECTION SYSTEM

4.2.1 Facilities shall be designed on the basis that city fire water supply is not available close to the installation.

4.2.2 Fire protection system shall be provided based on aggregate storage capacity as per below mentioned table:

Sl. No	Aggregate storage capacity of the Location	Fire Contingency	Fire Water Demand calculation criteria
1	For installation having aggregate underground and/or semi buried storage tank capacity up to 5000KL (Class A,		a. First aid Firefighting equipment consisting of portable and wheeled fire extinguishers, emergency trolley and emergency kit (as per Clause No. 4.7) b. Fire protection measures shall be implemented based on the risk identified



	B & C).		in the HIRA study or consequence analysis, duly approved by the competent authority of entity.
2	For installation having aggregate storage tank capacity up to 1000 KL (Class A, B & C) containing above ground tanks.	Single fire contingency	- Fire water flow rate shall be based on operating 4 single hydrant outlets and 1 water monitors simultaneously, capacity of each hydrant outlet shall be 36 m³/hr and of monitor shall be 144 m³/hr. - The effective capacity of the Fire water storage tank above the level of suction point shall be sufficient for minimum 1 hr of firefighting at designed fire water flow rate.
3	For installations aggregate above ground storage tank capacity up to 5000 KL (Only Class C & excluded petroleum product's).	Single Contingency Fire	- First aid Firefighting equipment consisting of portable and wheeled fire extinguishers, emergency trolley and emergency kit (as per Clause No. 4.7) - Fire protection measures based on risk identified in HIRA study or consequence analysis.
4	For installations aggregate storage tank capacity above 5000 KL (Only Class C & excluded petroleum product's).	Single contingency fire	- Fire Water Flow rate shall be based on using 4 single hydrant outlets and one water monitors simultaneously; capacity of each hydrant outlet shall be 36 KL/hr and of monitor shall be 144 KL/hr. - Effective capacity of the Fire water storage tank above the level of suction point shall be sufficient for minimum 4 hrs of firefighting at designed fire water flow rate. - Fire protection measures based on risk identified in HIRA study or consequence analysis.
5	For Installation Having aggregate storage tank capacity above 1000 kl & up to 30,000 kl (Including storage of Class C & excluded petroleum products , if stored with Class A	Single fire contingency	- Fire Water Flow rate required for largest Floating roof tank protection: - Cooling Water is required for the largest tank on fire. (Clause 4.3.1.1. (i). - Cooling water required for nearby tanks located in the same dyke enclosure. (Clause 4.3.1.1. (ii)). - Water flow required for applying foam on a largest floating roof tank burning surface area (rim-seal area). Clause 4.4.8)



	and/or Class B)	Total water flow rate required for largest floating roof tank protection = 1.a + 1.b + 1.c. 2. Fire Water flow rate required for largest Cone roof tank protection: a. Cooling Water required for the largest cone roof tank on fire. (Clause 4.3.1.1. (i)). b. Cooling water required for nearby tanks located in the same dyke enclosure. (Clause 4.3.1.1. (i)). c. Water flow required for applying foam on a largest cone roof tank burning surface area. (Clause 4.4.8) Total water flow rate required for largest cone roof tank protection = 2.a + 2.b + 2.c. 3. Fire water flow rate required for cooling POL tank wagon loading gantry (refer; Clause 4.3.1.1. (ii)). 4. Fire water flow rate required for full surface fire on largest floating roof tank (roof sinking case) a. Cooling Water required for the largest Floating roof tank on fire. (Clause 4.3.1.1. (i)). b. Cooling water required for nearby tanks located in the same dyke enclosure. (Clause 4.3.1.1. (i)). c. Water flow required for applying foam on a largest floating roof tank for full surface fire (roof sinking case) (Clause 4.4.8) Total water flow rate required for full surface fire on largest floating roof tank (Roof sinking case) = 4.a + 4.b + 4.c. Total design water flow rate for single fire contingency will be the highest of one of the fire water flow rate requirements calculated in 2, 3 or 4 + supplementary water (36 X 4 = 144 m³/hr) The number of main fire pumps shall be calculated based on total design water flow rate for single largest contingency. Fire water storage requirement for four hours of firefighting shall be based on aggregate capacity of main fire pumps (excluding standby pumps).
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**FIRE PROTECTION FACILITIES FOR PETROLEUM
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6	For Installation Having aggregate storage tank capacity above 30,000KL (Including storage of Class C products if stored with Class A and/or Class B)	Two Major Fires simultaneously (Double fire contingency)	1. Fire Water Flow rate required for largest cone or floating roof tank protection (considering dyke 1 on fire): a. Total water flow rate required for largest cone or floating roof tank protection shall be calculated as per above mentioned clause 5. 2. Fire Water flow rate required for second largest cone or floating roof tank protection (Considering dyke 2 on fire): a. Cooling Water required for the largest cone or floating roof tank protection shall be calculated as per above mentioned clause 5. 3. Fire water flow rate required for cooling POL tank wagon loading gantry (refer; Clause 4.3.1.1. (ii)). 4. Fire water flow rate required for full surface fire on largest floating roof tank (roof sinking case) a. Cooling Water required for the largest Floating roof tank on fire. (Clause 4.3.1.1. (i)). b. Cooling water required for nearby tanks located in the same dyke enclosure. (Clause 4.3.1.1. (i)). c. Water flow required for applying foam on a largest floating roof tank for full surface fire (roof sinking case) (Clause 4.4.8) d. Total water flow rate required for full surface fire on largest floating roof tank (Roof sinking case) = 4.a + 4.b + 4.c. 5. Total design water flow rate for two largest (double) fire contingency will be either the sum of two largest fire water flow rate requirements calculated in 6.1, 6.2 & 6.3 of this table + supplementary water ($36 \times 4 = 144$ cum/hr) or water requirement for single largest floating roof tank roof sinking scenario 6.4 + supplementary water ($36 \times 4 = 144$ cum/hr). 6. The number of main fire water pumps shall be calculated to meet the maximum fire water demand based on two (double) largest fire contingencies.
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			7. The fire water demand shall be determined considering 65 minutes (additional 30 minutes if foam stored for extended period) of foam application and 4 hours of tank cooling (including tank on fire) in the same dyke enclosure with supplementary water requirements for four hours .
7	For installation having aggregate storage tank capacity above 1,00,000KL (Including storage of Class C products if stored with Class A and/or Class B separation distances)	Two Major Fires simultaneously (Double fire contingency)	Requirements shall be calculated as per the above-mentioned clause 6. Only fire water for supplementary hose stream based on 4 hydrant streams + 1 High-Volume Long-Range water monitor. (i.e. $4 \times 36 \text{ m}^3/\text{hr} + 1 \times 228 \text{ m}^3/\text{hr} = 372 \text{ m}^3/\text{hr}$)

Note:- Fire water provision for supplementary hose stream shall be considered in new projects, based on four (4) hydrant streams + one (1) high-volume long-range water monitor.

4.2.3 Tank Wagon (TW)/Tank Truck (TT) loading/unloading facilities, Manifold area of product pump house & exchange pit shall be fully covered with a well laid out combination of hydrants and variable flow type water-cum-foam monitors. The components of the system shall be listed or approved by national or international agencies such as BIS, UL, and FM etc.

4.2.4 TW loading gantries shall be provided with fixed water spray system. The gantry shall be divided into suitable number of segments (each segment having maximum length of 15 m length & width of 12 m) and three largest segments operating at a time shall be considered as single risk for calculating the water requirement.

A provision shall be made to actuate the water spray system from a safe approachable location.

4.2.5 Fixed Water Spray on storage Tanks

- i) Class 'A' Petroleum storage in above ground tanks shall have fixed water spray system, whether floating roof or fixed roof having diameter larger than 6 meters.
- ii) Class 'B' above ground Petroleum storage tanks (fixed roof or floating roof) of diameter larger than 30 m shall be provided with fixed water spray system.

When Class A & B above ground storage tanks are placed in a common dyke, the fixed water spray system shall be provided on all tanks having diameter larger than 6 meters.

Note : Entity shall ensure cooling of Class 'B' above ground petroleum storage tanks (fixed roof or floating roof) of diameter less than 30 m.

4.2.6 The fixed water spray system shall also be provided on all tanks, irrespective of diameter

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in the installations, if inter distances between tanks in a dyke and/or within dykes are not meeting the requirements of OISD-STD-118.

4.2.7 Semi-fixed foam system or fixed foam system shall be provided for the following tanks:
Floating roof tanks storing Class 'A' and Class 'B' petroleum products.

- Fixed roof tanks storing Class 'A' and class 'B' petroleum products, of diameter ≥ 9 m.
- Fixed roof tanks storing class 'C' petroleum products, of diameter larger than 40 M.

4.2.8 Portable foam and/or variable flow water-cum-foam monitors shall be provided for suppression of pool fire in tank farm area. The components of the system shall be listed or approved by national or international agencies such as BIS, UL and FM etc.

4.2.9 An automatic actuated Rim Seal fire detection and extinguishing system shall be provided with the following requirements on all external floating roof tanks storing Class A Petroleum products.

Detection System: Linear heat detection system like hollow metallic tube type or two wired detection system or equivalent linear heat detection system.

Extinguishing Media: Foam based extinguishing media.

The major components of the system such as detection system, electrical / electronic components, foam discharge devices etc. shall be listed or approved by national or international agencies such as BIS, UL, and FM etc.

Note:- The use of equivalent or superior performance systems / methods to those already prescribed in the standard, should be considered, provided these equivalent systems / methods are duly approved by a technical committee constituted by OISD, based upon technical documentation, performance record and field demonstration by the user industry.

4.2.10 Fixed water spray system shall also be provided in lube oil drum areas if located in hazardous area.

4.2.11 CLEAN AGENT

Clean Agent based fire suppression system should be provided in Control Room, including Satellite Rack Room (SRR), engineering Room, computer rooms/ repeater station and pressurized rooms in major locations having automated pipeline receipt/dispatch and/or TW/TT loading facilities.

Selection of Clean Agent and design of Fire protection system shall follow the Standard on "Clean Agent Extinguishing systems NFPA Standard 2001 (latest edition) or equivalent including its safety guidelines with respect to "Hazards to Personnel", electrical clearance and environmental factors in line with environmental considerations of Kyoto & Montreal protocols and latest MoEFCC guidelines.

The gas-based (clean agent) extinguishing system should be provided in control rooms which are manned continuously (24x7 basis) and where a suitably designed smoke detection and alarm system is available. Clean Agent System along with all equipment, software for hydraulic calculations and corresponding design concentration of clean agent shall be listed or approved by national or international standards such as BIS/UL/FM/VdS/LPCB etc. Cylinder along with cylinder valve assembly shall have regulatory approval.

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4.2.12 COMBINED POL AND LPG FACILITIES IN THE SAME PREMISES

The fire protection and firefighting requirement for the combined POL and LPG facilities in the same premises shall be based on the following: -

- i. Each POL / LPG facility shall independently meet the design, layout & fire protection system requirements of corresponding OISD standards.
- ii. The fire water requirement shall be based on two fire contingencies simultaneously in the combined facility and fire water storage capacity shall be fixed accordingly.
- iii. The fire water system shall ensure availability of pressure of 7 kg/cm²g at the farthest point.
- iv. The entire fire water system shall remain pressurized and kept in auto mode.
- v. The control of such facility shall be clearly defined.

4.3 FIRE WATER SYSTEM

Water is used for fire extinguishment, fire control, cooling of equipment, exposure protection of equipment and personnel from heat radiation. The fire water ring main shall be provided all around the installation.

4.3.1 COMPONENTS OF FIRE WATER SYSTEM

The main components of the system are Fire Water Storage, Fire Water Pumps and Distribution Piping Network.

4.3.1.1 DESIGN FLOW RATE

- i) Fire water flow rate for a tank farm shall be aggregate of the following: -
 - a. Water flow calculated for cooling a tank-on-fire at a rate of 3 lpm/m² of tank shell area.
 - b. Water flow calculated for all other tanks situated in the same dyke area at a rate of 3 lpm/m² of tank shell area + roof (in case of Fixed roof tank).
 - c. Water flow required for applying foam into a single largest cone roof or floating roof tank (Rim seal area) burning surface area of oil, by way of fixed/semifixed system, where provided or by use of foam monitors. (Refer clause 4.4.8 of this standard).
 - d. Water flow required for applying foam into a single largest cone roof or floating roof tank (after the roof has sunk) burning surface area of oil, by way of fixed/semi fixed foam system, where provided or by use of water cum foam monitors. (Refer clause 4.4.8 of this standard)
 - e. For water flow calculations, all tanks farms having class A or B petroleum storage shall be considered irrespective of diameter of tanks and whether fixed water spray system is provided or not.

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- f. Water flow required for applying foam on a single largest tank by way of fixed foam system, where provided, or by use of water/foam monitors. (Refer section 4.4.8 for foam solution application rates.)
- g. Various combinations shall be considered in the tank farm for arriving at different fire water flow rate and the largest rate to be considered for design.
- ii) Fire water flow rate for TW loading gantry (Refer Annexure-II) and product pump house in a depot or terminal shall be calculated at a rate of @ 10.2 lpm/m².
- iii) For Pump of volatile product/s located under pipe rack fire water flow rate shall be calculated at a rate of 20.4 lpm/ m².
- iv) Fire water flow rate for supplementary streams shall be based on using 4 single hydrant outlets. Capacity of each hydrant outlet as 36 m³/hr shall be considered at a pressure of 7 kg/cm² (g). For installation having aggregate storage tank capacity above 1,00,000KL, fire water flow rate for supplementary streams shall be based on using 4 single hydrant + 1 High-Volume Long-Range water monitor. (i.e., 4 x 36 m³/hr + 1 x 228 m³/hr = 372 m³/hr). The supplementary water stream requirement shall be in addition to design flow rates as calculated in (i), (ii), (iii) & (iv) above.
- v) The design fire water rate shall be aggregate of the largest of 4.3.1.1 {(i),(ii), or (iii)] and 4.3.1.1(iv). (Refer Annexure-I).

4.3.2 HEADER PRESSURE

Fire water system shall be designed for a minimum residual pressure of 7 kg/cm² (g) at hydraulically remotest point of the installation, at the corresponding design flow rate, while accounting for either single or double fire contingency scenarios, as applicable.

4.3.3 STORAGE

Water for the firefighting shall be stored in easily accessible underground or above ground tanks of steel, concrete, or masonry.

The effective capacity of the reservoir/tank above the level of suction point shall be adequate to meet the calculated fire water demand. The fire water demand shall be determined considering 65 minutes (additional 30 minutes if foam stored for extended period) of foam application and 4 hours of tank cooling (including tank on fire) in the same dyke enclosure with supplementary water requirements for four hours excluding for installation having storage tank capacity up to 1000 kl (Class A, B & C) containing above ground tanks where it shall be for 1 hour. Main fire water pumps shall be designed to meet the maximum fire water demand requirement.

For all locations, where make up water replenishment at the rate of 50% or more is available, the fire water storage capacity can be reduced to 3 hours of the aggregate rated capacity of pumps considering exclusion as above.

Fire water should be used only for firefighting purposes. In case sea water or treated effluent water is used for firefighting purposes, the material of the pipe selected shall be suitable for the service.

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The installation shall have facilities for receiving and diverting the supply water, coming to the installation, to fire water storage tanks, in case of an emergency.

Large natural reservoirs having water capacity exceeding 10 times the aggregate fire water requirement can be left unlined.

4.3.4 FIRE WATER PUMPS

- (i) Fire water pumps having flooded suction shall be installed to meet the design fire water flow rate and head.
- (ii) The pumps shall be capable of discharging 150% of its rated discharge at a minimum 65% of the rated head. The Shut-off head shall not exceed 120% of rated head for horizontal centrifugal pumps and 140% for vertical turbine pump.
- (iii) At least one standby fire water pump shall be provided up to 2 nos. of main pumps. For main pumps 3 and above, minimum 2 standby pumps of the same type, capacity & head as the main pumps shall be provided.
- (iv) The fire water pump(s) including the standby pump(s) shall be of diesel engine driven type. The diesel engines shall be quick starting type with the help of push buttons located on or near the pumps or located at a remote location. Each engine shall have an independent fuel tank adequately sized for 6 hours continuous running of the pump.
- (v) Fire water pumps & storage shall be located as per OISD-STD-118.
- (vi) Fire water pumps shall be exclusively used for firefighting purpose only.
- (vii) Suction and discharge valves of fire water pumps shall be kept full open all the times.
- (viii) The fire water network shall be kept pressurized by jockey pump(s).
- (ix) The jockey pump shall be sized adequately to compensate for the pressure drop in the fire water header arising from system leakage and normal pressure losses and shall maintain the fire water header pressure at 7.0 kg/cm².
- (x) The fire water pumps shall be provided with auto start facility with pressure drop in fire water network.

4.3.5 FIRE WATER NETWORK

(i) LOOPING

The fire water network shall be laid in closed loops as far as possible to ensure multi-directional flow in the system. Isolation valves shall be provided in the network to enable isolation of any section of the network without affecting the flow in the rest. The isolation valves shall be located normally near the loop junctions. Additional valves shall be provided in the segments where the length of the segment exceeds 300 m.

(ii) ABOVE / UNDERGROUND NETWORK

The fire water network piping should normally be laid above ground at a height of at least

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300 mm above finished ground level. Fire water mains shall not pass through buildings or dyke areas, foundations of equipment or structures. Pipes made of composite material shall be laid underground.

The ring main shall be laid underground at the following places.

- Road crossings.
- Places where above ground piping is likely to cause obstruction to operation and vehicle movement.
- Places where above ground piping is likely to get damaged mechanically.
- Where frost conditions warrants, and ambient temperature is likely to fall sub-zero, above ground piping shall be laid at least 1 m below the finished grade level to avoid freezing of water. Alternatively, water circulation should be carried out in the above ground pipelines or any other suitable means.

(iii) PROTECTION OF UNDERGROUND PIPELINE

If fire water ring mains are laid underground, the following shall be ensured: -

- The ring main shall have at least 1 m earth cushion in open ground, 1.5 m cushion under the road crossings and in case of crane movement area pipeline shall be protected with concrete/steel encasement as per design requirement.
- For rail crossing, provisions stipulated by Indian Railways shall be complied. The underground ring main shall be protected against soil corrosion by suitable coating/wrapping with or without cathodic protection. Where pipes are coming out/going in from/into the ground, the underground anticorrosion coating on the pipe will continue for a length of at least 300 mm above ground.
- Pipe supports under the pipeline shall be suitable for the soil conditions.

(iv) SUPPORT & PROTECTION OF ABOVE GROUND PIPELINES

The mains shall be supported at regular intervals not exceeding 6 m. For pipeline size less than 150 mm, support interval shall not exceed 3 m. The pipe support shall have only point contact.

The system for above ground portion shall be analysed for flexibility against thermal expansion and necessary expansion loops, guides/cross guides and supports provided.

(v) SIZING OF PIPELINE

- a) The hydraulic analysis of network shall be done at the design time. Also, whenever fire water demand increases due to addition of facilities or extensive extension of network, fresh hydraulic analysis shall be carried out.
- b) Fire water ring main shall be sized for 120% of the design water flow rate. Velocity of the water shall not exceed 5m/s in the fire water ring main. Design flow rates shall be distributed at nodal points to give the most realistic way of water requirements in an emergency. It may be necessary to assume several combinations of flow requirement for

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design of network.

(vi) GENERAL

Connections for fixed water Monitors and Hydrant posts on the network shall be provided with independent isolation valves.

In case of underground mains, the isolation valves shall be located in RCC/brick masonry chamber of suitable size to facilitate operation during emergency & maintenance.

Flushing connections with isolation valves should be provided at suitable locations in the firewater ring main. Pressure gauge/Pressure transmitter shall be provided at strategic locations to monitor the header pressure.

4.3.6 HYDRANTS & MONITORS

- i) At least one hydrant post shall be provided for every 30 m of external wall measurement or perimeter of battery limit in case of hazardous areas. For non-hazardous area, they shall be spaced at 45 m intervals. The horizontal range & coverage of hydrants with hose connections shall not be considered beyond 45 m.
- ii) Hydrants/Monitors/HVLRM shall be located at a minimum distance of 15 m from the periphery of storage tank or equipment under protection. In case of buildings this distance shall not be less than 2 m and not more than 15 m from the face of building. Provision of hydrants within the building shall be provided in accordance with IS: 3844.
- iii) Hydrant/Monitors shall be located along roadside berms for easy accessibility.
- iv) A two-way hydrant shall be used. All hydrant outlets shall be situated at workable height of not exceeding of 1.2 meter above ground or hydrant/monitor operating platform level.
- v) Multihydrant outlet (Hydrant manifold with minimum 4 outlets) shall be provided at suitable locations for feed to fire tender or portable firefighting equipment.
- vi) Monitors shall be located to direct water on the object to be protected. The requirement of monitors shall be established based on hazards involved and layout considerations. Monitors shall not be installed within 15 m of equipment to be protected.

In pipeline installation minimum two numbers of water cum foam monitors shall be located in such a way that it covers the pump area, scrapper area and filter area.

The location of monitors shall not exceed the performance range of the monitor from the object to be protected.

- vii) For marketing terminals & Depots, the remote operated high volume long range water cum foam monitors to fight tank fires shall be provided. Monitor shall be capable of inducing foam from distance up to 60 m.

For Pipeline installations and Central Tank Farms, Manual or / and Remote operated high-volume long-range water cum foam monitors (Fixed/Trolley type) to fight tank fires shall be provided. Manual type HVLR shall be considered only where sufficient distance (15 meter) is available from the point of hazards and in-company firefighting infrastructure (round the clock fire tender and dedicated manpower) like in Refineries with respect to the location is available.

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The location of HVLRs to be planned in such a way that the very purpose of these monitors is served and throw of the monitors is safely delivered at the aimed object.

The foam tank and other accessories for induction of foam to the HVLR monitor shall be positioned at a safe location for easy access and replenishment of foam in emergency. However, remote control panel for HVLR monitor should be at located beyond radiation contour of 4 kw/m² from the tank on fire and the projectile of the foam & target object must be visible to the operator.

The basic water-cum-foam monitors shall be listed or approved by national or international standards such as BIS, UL, and FM. The electrical or hydraulic remote-control mechanism shall be in line with Hazardous Area Classification.

Following criteria shall be followed for installation of water-cum- foam monitors:

- a) Remote or / and manual operated monitors shall be installed in such a way that all the tanks in the installation are within its performance range.
- b) Minimum two nos. of fixed type HVLR monitors shall be installed for each tank containing Class A in the installation to meet the requirement as per s.no. (a) above. Monitor/s shall be provided to meet foam application rate of 9.75 lpm/m². Monitors shall be located considering prevailing wind direction and distance from the tank to achieve the desired outcome.
- c) For all existing storage tanks not meeting the safety distance norms as per OISD-STD-118, below requirement shall be followed.
 - I. HVLR water cum foam monitors of fixed type shall be provided for the tanks storing Class B&C products also. Requirement of HVLR monitors shall be calculated for full surface fire scenario of the largest tank @9.75 lpm/m². In respect of installation storing exclusively Class-C product this provision shall not be applicable.
 - II. In addition to the above, the entity shall carry out a Quantitative Risk Assessment (QRA) for facilities that do not comply with the inter-distance norms for storage tanks, in line with the provisions of OISD-STD-118. Based on the QRA findings, the entity shall implement recommended mitigation measures.
 - III. Hydrants and monitors shall not be installed inside the dyke areas However, in exceptional circumstances, if required, remote operated elevated monitors shall be provided inside the dyke area to cover the inaccessible portion of tanks which is not getting covered through external monitors. The monitor shall be provided with isolation valve or ROV outside the tank farm. The monitor shall be installed on an independent elevated structure with a minimum height exceeding the dyke height. The supporting structure shall be fireproofed for pool fire scenario. The riser pipe to monitor shall also be fireproofed. The Hydrant pipeline inside the dyke area shall be underground. In case, above ground hydrant pipeline is provided inside the dyke area, the same shall be fireproofed. The remote-control panel for HVLR monitor shall be at a distance of minimum of 60 m away from the vessels to be protected.

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- d) At small installations having aggregate storage capacity up to 5000kl, where it is not possible to install the fixed type HVLR monitors at a safe distance (minimum 15 m) from the tank because of non-availability of adequate space/distance, shall be allowed to provide trolley mounted monitor of suitable capacity for Class A&B tank also.
- e) Provision for connecting / hooking the portable monitor shall be made in the hydrant system around storage tanks at various strategic points.
- viii) TW/TT loading & unloading facilities shall be provided with alternate hydrant and water-cum-foam monitors (BIS/UL/FM listed/approved) having multipurpose combination nozzles for jet, spray & fog arrangement and located at a spacing of 30 m on both sides of the gantry. The hydrants & monitors shall be located at a minimum distance of 15 m from the hazard (e.g. TW & TT loading/unloading facilities) to be protected.
- ix) Hydrants / Monitors shall be located with branch connections and not directly over main header for easy accessibility and to avoid corrosion of main header due to dripping from hydrants/accumulation of water, if any.

4.3.7 MATERIAL SPECIFICATIONS

All the materials required for a firewater system using fresh water shall be of approved type as indicated below and in line with OISD-GDN-115.

i) PIPES

Carbon Steel as per IS: 3589/IS: 1239/IS: 1978 or Composite Material or its equivalent for freshwater service.

In case saline, blackish or treated effluent water is used, the fire water ring main of steel pipes, internally cement mortar lined or glass reinforced epoxy coated or pipes made of material suitable for the quality of water shall be used. The material selection for the fire main line shall consider the quality or impurities in the water. Alternately, pipes made of composite materials shall be used.

The composite material to be used shall be as per API 15LR/API 15HR/API 15S.

ii) ISOLATION VALVES

Cast Steel valves shall be used in all areas including unit areas, offsite and fire water pump stations. Isolation valves shall be Gate valves having open/close indication. More than 12" valves should be provided with gear mechanism.

4.3.8 FIXED WATER SPRAY SYSTEM

- i) Fixed water spray system is a fixed pipe system connected to a reliable source of water supply and equipped with water spray nozzles for specific water discharge and distribution over the surface of area to be protected. The piping system is connected to the Fire water network through an automatically or manually actuated valve which initiates the flow of water.

In case the system is manually actuated, the isolation valve shall be located outside the dyke for ease of access & operation.

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- ii) Spray nozzles shall be directed radially to the tank at a distance not exceeding 0.6 m from the tank surface.
- iii) While calculating the water rates for spray application for cases other than tanks/vessels, the area should be divided into suitable segments so that maximum water requirement can be optimized. (Refer Annexure-II for typical calculations).

4.4 FOAM SYSTEMS

For combating large hydrocarbon fires particularly in a contained area like storage tank, foam has proved useful for its inherent blanketing ability, heat resistance and security against burn-back. Efficient and effective foam delivery system is a vital tool for its usefulness in controlling fire. The process of adding or injecting the foam to water is called proportioning. The mixture of water and foam compound (foam solution) is then mixed with air in a foam maker for onward transmission to burning surface.

4.4.1 TYPES OF FOAM

Foams are classified by producing action of generation and expansion. Foam concentrates to be used shall conform to IS: 4989 /UL-162 or equivalent Standard (Annexure – IV)

4.4.2 CONVEYING SYSTEMS

The system consists of an adequate water supply, supply of foam concentrate, suitable proportioning equipment, a proper piping system, foam makers and discharge devices designed to adequately distribute the foam over the hazard.

Conventional systems are of the open outlet type, in which foam discharges from all foam outlets at the same time, covering the entire hazard within the confines of the system.

There are three types of systems: -

- i) Fixed
- ii) Semi-Fixed
- iii) Mobile

(i) FIXED FOAM SYSTEM

Fixed foam conveying system comprises of fixed piping for water supply at adequate pressure, foam concentrate tank, eductor, suitable proportioning equipment for drawing foam concentrate and making foam solution, fixed piping system for onward conveying to foam makers for making foam, vapor seal box and foam pourer. System operation may be automatic for immediate detection and actuation for the foam suppression for minimum duration as defined in case of floating roof and fixed roof tanks or manual in operation.

(ii) SEMI-FIXED FOAM SYSTEM

Semi-fixed foam system gets supply of foam solution through the mobile foam tender. A fixed piping system connected to foam makers cum vapor seal box in case of cone roof tanks and foam maker and foam pourers in the case of floating roof tanks conveys foam to the burning surface of tank.

(iii) MOBILE SYSTEM

Mobile system includes foam producing unit mounted on wheels which should be self-propelled or towed by a vehicle. These units supply foam through monitors/foam towers to

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the burning surface.

4.4.3 EXTERNAL FLOATING ROOF TANK PROTECTION

For floating roof tank, foam shall be poured at the foam dam to blanket the roof seal. Features of foam system for floating roof tank protection shall be as follows: -

4.4.3.1 PROTECTION USING AUTOMATIC FIRE DETECTION AND FOAM SUPPRESSION SYSTEM:

Automatic actuated Rim Seal fire detection and extinguishing system shall be provided on all external floating roof tank storing Class A Petroleum products. The components of the system shall be listed or approved by national or international agencies such as BIS, UL and FM etc. The minimum requirement of design for the automatic actuated foam flooding system is given in Annexure V.

4.4.3.2 PROTECTION USING FIXED / SEMI-FIXED FOAM SYSTEM

- i. Fixed Foam suppression system should be fully automatic comprising foam tank, inline induction, foam proportioning and fixed piping to be operated on automatic detection of fire at rim seal area.
- ii. In case of Semi-Fixed foam system, foam inlet breeching connections shall be located at accessible location.
- iii. System shall be designed to create foam blanket on the burning surface in a reasonably short period.
- iv. Continuous foam concentrate supply shall be applied to the burning hazard at a rate high enough to overcome the destructive effects of radiant heat by Centralized foam system / multiple foam nurseries.
- v. Foam makers/foam pourers shall be located not more than 24 M apart on the shell perimeter based on 600 mm foam dam height. The height of foam dam shall be at least 51 mm above the top of metallic secondary seal.
- vi. A minimum of two foam pourers shall be provided.
- vii. Whenever mobile type of systems like foam nurser are used for supplying foam to HVLR of storage tank, provision shall be made to connect more than one foam nurser for continuous foam concentrate supply.

4.4.4 FIXED ROOF TANK PROTECTION

Foam conveying system shall have same features as of floating roof tank excepting that a vapor seal chamber is required before the foam discharge outlet.

Features of the foam system for fixed roof protection shall be as follows:

- i) The vapor seal chamber shall be provided with an effective and durable seal, fragile under low pressure, to prevent entrance of vapor into the foam conveying piping system.
- ii) Where two or more pourers are required, these shall be equally spaced at the periphery of the tank, and each discharge outlet shall be sized to deliver foam at approximately the same rate.

iii) Tanks should be provided with foam discharge outlets/pourers as indicated below :-

Tank Diameter in m. Minimum number of foam discharge outlet

> 9 upto 20	2
>20 upto 25	3
>25 upto 30	4
>30 upto 35	5
>35 upto 40	6
>40 upto 45	8
>45 upto 50	10

Plus 1 outlet for

each additional 465 m² of area

The estimation of number of foam discharge outlet is based on maximum pourer capacity of 1000 lpm at a pressure of 7 kg/cm² (g) upstream of eductor. This can be suitably adjusted for different pourer capacity in accordance with section 4.4.4.3 (v).

4.4.5 FLOATING CUM FIXED ROOF TANK PROTECTION

Protection facilities shall be provided as per clause 4.4.4.

4.4.6 FULL SURFACE TANK FIREFIGHTING & LOGISTICS

Large Diameter Full Surface Tank Fire Fighting:

Full surface Tank Fire Scenario:

Part A: Tanks diameter up to 45 meter

It is experienced that full surface fire in smaller diameter petroleum storage tank up to 45-meter diameter are relatively easy to extinguish with reliable resources and competent manpower.

This shall be achieved by designing dyke foam monitors of adequate capacity / foam tender roof monitors. Un-interrupted foam supply shall be ensured by any reliable means for a continuous period of 65 minutes.

Part B: Tanks diameter larger than 45 meters

Large diameter full surface tank fire is a challenge and requires significant preparation in terms of resources and well proven strategy.

Successful extinguishment of a large diameter petroleum storage tank full surface fire begins with pre-fire planning and therefore most of the following information shall be considered. Entity to prepare the pre-fire plan for largest diameter tank at site and list out the resources required. The priority shall be for rescue and evacuation of personnel before attempting extinguishment.

The manpower requirement to tackle a major tank fire may vary depending upon the type,

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location and nature of the fire, method of extinguishment and the availability of trained personnel. This shall be determined while developing the pre-fire plans.

Resources Requirement:

Below listed major resources shall be made available at site as a design requirement and periodically these resources should be tested and maintained in ready to use condition. The critical aspect of successful tank fire extinguishment is a stable uninterrupted foam delivery in required quantity through aerial (over the top) application. Foam discharge through large volume monitors should reach close to tank center for effective foam blanketing over burning surface. Many successful full surface tank fire fighting are achieved by throwing foam through 2-3 large volume foam monitors from upwind side of the burning tank. Multiple direction foam discharge over a burning tank may not give a satisfactory result.

A typical list of major Trolley /portable / mobile equipment for full surface tank fire is given below.

- a. High-Volume Long-Range foam monitors
- b. Water monitors
- c. Foam pumping/ proportioning appliances
- d. Foam concentrate tankers or containers (Tote containers)
- e. Fire hose including large diameter/ capacity hose and Water supplies.
 - a. High Volume Long Range (HVLR) Foam Monitors

The foam monitor capacity and range are important factors for fire extinguishment. Foam streams have to be such that the bulk of the discharge reaches the tank liquid surface. It is recognized that the best method of application is to project foam with the wind behind the stream and not against. However, there have been, and will be situations where cross winds or a variable breeze causes destructions in stream range and these needs to be considered. Therefore, it is important to use large capacity trailer mobile foam monitors for tank fire extinguishment, which give the versatility for placement considering the prevailing wind direction so that the bulk of the output reaches the tank liquid surface without affecting the stream range. Number and capacity of large volume foam monitors should be decided based on Pre-fire plan for largest diameter tank fire. Foam monitors are available ranging from 1000 GPM (228 m³/hr) to 12000GPM (2736 m³/hr).

b. Water Monitors

Water monitors are necessary for cooling adjacent tanks affected by radiant heat or direct flame impingement. Water cooling of the tank on fire is not normally necessary except to assist foam blanket sealing against a hot tank wall. Cooling may be attempted if the water spray system is intact and operable for uniform cooling.

c. Foam proportioning appliances.

An uninterrupted foam supply with required induction rate matching with foam monitors discharge for the total duration of foam application is necessary for extinguishment. It is very essential that the selected foam proportioning system is suitable for the logistical challenges involved in bulk foam concentrate handling. It is possible to supply foam to the foam monitors

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either using foam pumpers, or foam fire tenders or through foam pick up of a monitor.

d. Foam concentrate and supply considerations.

Availability of required quantity of foam concentrate at emergency site shall be ensured before commencing foam application. Once foam application commences, it shall be sustained for a minimum period of 65 minutes @ 9.75 lpm/m². It is recommended that bulk storage is provided in foam nurser trucks, elevated bulk storage tanks or 1000-liter Tote containers on mobile trailers / trucks. Conventionally 3% foam concentrates are used for tank fire fighting, however 1% foam concentrate can be considered for better logistics management. It is highly recommended to keep the total quantity of foam as a separate stock to be mobilized only during tank fire scenario.

It is experienced that foam losses during actual firefighting is high around 50-60 % during heavy wind or adverse weather condition or flame draft from the burning tank. Safety allowance of additional 50% has been included in the above application rate.

Foam compatibility is an important factor. Mixing foam concentrates of different types is not recommended and can destroy foam blanket over the burning surface. Moreover, foam concentrate should be compatible with associated resources being used for firefighting.

Suggested options for Foam concentrate supply-

- Foam Nurser / Foam Transport Vehicle / Foam Tender
- Foam Tote on Trailer with common discharge manifold

e. Fire Hose and Water supply.

Wherever the full surface firefighting strategy is adopted with mobile trailer mounted high-capacity water cum foam monitors, large capacity hoses for feeding water to the monitor should be selected to avoid losses / pressure drop during actual operation. The sizes of large capacity hoses vary from 4" to 12". Conventional 63mm fire hoses may be used for portable water monitors for cooling and foam proportioning system. Power driven device may be used for quick laying of hoses.

Fire water from available fire water network / large capacity pumps should be used to feed large diameter hoses for monitors. Suitable feeding arrangement should be made available for this water supplies.

4.4.7 PROTECTION FOR DYKE AREA/SPILL FIRE

Portable monitors/foam hose streams shall be provided for fighting fires in dyked area and spills. Additionally, Medium expansion foam generators shall be provided to arrest vapor cloud formation from spilled volatile hydrocarbons.

Installation of medium expansion foam generator shall be as per following criteria:

Class A tanks:

Minimum 2 nos. Fixed type foam generators for each tank dyke.

Class B tanks:

Two nos. portable foam generator for each location.

**4.4.8 FOAM APPLICATION RATE**

The minimum delivery rate for primary protection based on the assumption that all the foam reaches the area being protected shall be as indicated below. In determining total solution flow requirements, potential foam losses from wind and other factors shall be considered.

For cone roof tanks containing liquid hydrocarbons, the foam solution delivery rate shall be at least 5 lpm/ m² of liquid surface area of the tank to be protected.

For floating roof tanks containing liquid hydrocarbons foam solution delivery rate shall be at least 12 lpm/ m² of seal area with foam dam height of 600 mm of the tank to be protected. The height of foam dam shall be at least 51 mm above the top of metallic secondary seal.

In the case of Floating roof tank roof sinking, the application rate shall be considered as 9.75 lpm/ m².

4.4.9 DURATION OF FOAM DISCHARGE

The equipment shall be capable of providing primary protection at the specified delivery rates for the following minimum duration.

- i) Tanks containing Class 'A' & 'B' - 65 minutes.
- ii) Where the system's primary purpose is for spill fire protection - 30 minutes.
- iii) Tanks containing liquid hydrocarbons - Class 'C' Petroleum - 30 minutes.

Note: - It is advised to keep foam compound stock for extended 30 minutes for full surface tank fire scenario.

4.4.10 WATER FOR FOAM MAKING

Water quantity required for making foam solution depends on the percent concentration of foam compound. Foams in normal use have a 1%, 3% or 6% proportioning ratio. However, foam supplier data shall be used for determining water requirement.

4.4.11 FOAM QUANTITY REQUIREMENT

The aggregate quantity of foam solution should be calculated as below:-

- i) Foam solution application at the rate of 5 lpm/ m² for the liquid surface of the single largest fixed roof tank including cone roof/ internal floating roof or at the rate of 12 lpm/ m² of seal area of the single largest external floating roof tank whichever is higher. (Annexure-III).
- ii) In the case of Floating roof tank roof sinking, the application rate shall be considered as 9.75 lpm/ m².
- iii) Based on the size of the terminal, quantity of foam solution required should be calculated as per the following guidelines:-



Size of Terminal (In KL)	Water/Foam Monitor (Number)
For installation having Aggregate capacity of 1000KL	Minimum 1 number of mobile high-volume-long-range foam monitors having minimum capacity of 1000 GPM for locations having above ground tanks
For Installation having aggregate capacity above 1000KL & up to 30,000KL	Minimum 2 Numbers of mobile high-volume long-range foam monitors having minimum capacity of 1000 GPM.
For Installation having aggregate capacity above 30,000KL & up to 1,00,000KL	Minimum 2 Numbers of mobile high-volume long-range foam monitors having minimum capacity of 1500 GPM.
For Installation having aggregate capacity more than 1,00,000 KL	More than 2 Numbers of mobile high-volume long-range foam monitors having minimum capacity of 2000 GPM.

- iv) Two hose streams of foam each with a capacity of 900 lpm of foam solution.

The aggregate quantity of foam solutions should be largest of 4.4.12(i), 4.4.12(ii), 4.4.12(iii) and 4.4.12(iv) as above for a minimum period specified in Cl.4.4.9. From this the quantity of foam based on 1%, 3% or 6% proportion should be calculated.

Additional Foam quantity requirement & foam monitor requirement shall be in line with 4.3.6(vii).

4.4.12 FOAM COMPOUND STORAGE

- Foam compound should be stored as explained in IS-4989:2006/UL-162. Foam compound should be stored in containers of suitable capacity, in case of protein, fluoro protein or AFFF or AR-AFFF or FFFP.
- Alcohol Resistant Foam (AR-AFFF) shall be used for handling methanol/ ethanol fires.
- Alcohol Resistant Foam (AR-AFFF) can be used for all type of hydrocarbon fire and polar solvent fire effectively. Foam compound can also be stored in overhead storage tank of suitable capacity for quick filling of foam tender / nurer during emergency.

Cautionary: Alcohol Resistant Foam (AR-AFFF) is preferred for Ethanol Blended Petrol (EBP) fire.

Shelf life of foam compound shall be taken from manufacturer's data.

Foam compound shall be tested periodically in line with OISD-STD-142 to ensure its quality

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and the deteriorated quantity should be replaced. The deteriorated foam compound can be used for fire training purposes. For details of type of tests & their periodicity, refer IS 4989: /UL-162 or Equivalent Standard.

Quantity of foam compound equal to 100% of requirement as calculated in 4.4.11 shall be stored in the Installation. This quantity can be suitably reduced if mutual aid for foam supply is available. For sample calculation, refer (Annexure- III).

4.4.13 MODERNIZATION PLAN FOR FIREFIGHTING EQUIPMENT'S:

As a part of modernization of firefighting equipment, following may be explored in addition to the OISD-GDN-115 based on its suitability to hazards, long term maintainability, reliability, and certification.

a. Hydro-chem Nozzle

The nozzle is designed to throw liquid (foam solution / water) and DCP simultaneously. The liquid forms a hollow jet and the DCP is inside the hollow space. Application wise DCP provides quick knock down of flames and the liquid provides cooling. Also, DCP can be thrown at a longer distance than conventional DCP hand nozzle or monitor.

b. Fire extinguisher of HDPE / Composite material body: This will ensure less weight and longer shelf life. This is also included in IS 15683:2018.

4.5 FIREFIGHTING RELATED TO ETHANOL

- (i) For spill of ethanol at TLF gantry or during unloading of Ethanol TT alcohol resistant foam cover to be immediately spread over the spilled contents using portable Medium Expansion Foam Generator (MEFG) or appropriate equipment.
- (ii) For leak of ethanol from gaskets in Pump House or in Tank farm, alcohol resistant foam covers to be immediately spread over the spilled contents using portable MEFG or appropriate equipment.
- (iii) If in Tank Farm, common HVLR is used for tanks storing MS or ethanol or ethanol blended MS, then, AR-AFFF foam shall be used.
- (iv) Minimum two MEFG for each tank inside dyke area storing ethanol or ethanol blended MS shall be provided. Separate Alcohol Resistant Foam feeding arrangement shall be made for such MEFGs. Appropriate SOP shall be prepared and displayed near the operating area.
- (v) Hydrocarbon Detection system as applicable for ethanol shall be provided at all possible places of leak including isolation valve inside dyke and at sump near dyke drain valve.
- (vi) Foam pourer system as provided in Internal Floating Roof Vertical Tank (IFRVT) or in Cone Roof Vertical Tank (CRVT) for ethanol service shall have dedicated provision for Alcohol Resistant Foam for firefighting.

4.6 CONTROL ROOM AND COMPUTER ROOM PROTECTION

Fire Protection system to be provided for control and computer room shall comply with the requirements of OISD-STD-163 (Clause 6.0).

**4.7 FIRST AID FIRE FIGHTING EQUIPMENT****4.7.1 PORTABLE FIRE EXTINGUISHERS**

All fire extinguishers shall conform the requirements as specified in OISD-GDN-115.

- i) The maximum running distance to locate an extinguisher shall not exceed 15 m.
- ii) The extinguisher shall be installed in such a way that its top surface is not more than 1.5m above the floor/ground level.

The no. of extinguishers at various locations shall be provided as under.

Petroleum Depots, Terminals, CTF,CAVERNS & Lube Oil Installations

Sr. No.	Type of Area	Scale of Portable Fire Extinguishers
(i)	Lube Godown	1 No.9 DCP extinguisher for every 200 m ² or min. 2 Nos. in each Godown whichever is higher.
(ii)	Lube Filling Shed	1 No. 9-Kg DCP extinguisher for 200 m ² or min. 2 Nos. in each Shed whichever is higher
(iii)	Storage of (Class A/B) in packed containers and stored in open/closed area.	1 No. 9 Kg DCP extinguisher for 100 m ² or min. 2 Nos. in each Storage Area whichever is higher.
(iv)	Pump House (Class A/B) Up to 50 HP Above 50-100 HP Beyond 100 HP	Minimum-2nos 9 Kg DCP for 2 pumps. Minimum-2No. 9 Kg DCP for each pump. Minimum 2Nos. of 9 kg or 1 no. of 25 kg DCP for each pump.
(v)	Pump House (Class C) Up to 50 HP Above 50 HP	Minimum 2no. 9 Kg DCP for every 4 pumps up to 50 HP. 2 nos. 9 Kg DCP or 1x25 kg DCP for 4 pumps.
(vi)	Tank Truck loading & unloading gantry for POL/Special products	Minimum 2 No. 9 Kg DCP extinguisher for each bay plus 1 No. 75 Kg DCP extinguisher for each gantry.
(vii)	Tank Wagon loading and unloading gantry/siding	Minimum 2No. 9 Kg DCP extinguisher for every 30 m of gantry/siding plus 1 No. 75 Kg DCP extinguisher for each gantry/siding.
(viii)	A/G Tank Farm	2 Nos. 9 Kg DCP extinguishers for each tank plus 4 Nos. 25 Kg DCP extinguishers for each Tank Farm positioned at four corners. In case of adjoining tank farms, the no. of 25 Kg extinguishers can be reduced by 2 nos. per tank farm.
Sr. No.	Type of Area	Scale of Portable Fire Extinguishers
(ix)	Underground and/or semi buried storage tank	2 Nos. 9 Kg DCP extinguisher for each Tank Farm plus 1 No. 75 Kg DCP extinguisher
(x)	Other Pump Houses	1 No. 9 Kg DCP extinguisher for every two pumps or min 2 Nos. 9 Kg DCP extinguisher for each Pump House whichever is higher.
(xi)	Admin. Building/Store House	1 No. 9 Kg DCP extinguisher for every 200 m ² or min. 2 Nos. 9 Kg DCP extinguishers for each floor of Building/Store whichever is higher.



(xii)	DG Room	2 Nos. each 9 Kg DCP & 4.5 Kg CO ₂ extinguishers for each DG room.
(xiii)	Main switch Room/Sub-Station	1 No. 4.5 Kg CO ₂ extinguisher for every 25 m ² plus 1 No. 9 Liter sand bucket.
(xiv)	Portable Encapsulator agent extinguisher.	2 Nos. each 6L in battery rooms storing lithium-ion battery.
(xv)	Computer Room/ Cabin	2 Nos. of 2 Kg CO ₂ or 2 Nos. of 2.5 Kg Clean Agent extinguisher per Computer Room and 1 No. 2 Kg CO ₂ or 1 No. 1.0 Kg Clean Agent extinguisher per cabin.
(xvi)	Security Cabin	1 No. 9 Kg DCP extinguisher per cabin.
(xvii)	Canteen	1 No. 6 liter class F extinguisher for 100 m ² .
(xviii)	Workshop	1 No. 9 Kg DCP extinguisher & 1 No. 2 Kg CO ₂ extinguisher.
(xix)	Laboratory	1 No. 9 Kg DCP extinguisher & 1 No. 4.5 Kg CO ₂ extinguisher.
(xx)	Oil Sample Storage Room	1 No. 9 Kg DCP extinguisher per 100 m ² or min. 1 no. 9Kg extinguisher per room whichever is higher.
(xxi)	Effluent Treatment Plant/OWS	1 No. 75 Kg. & 2 nos. 9 Kg. DCP Extinguisher
(xxii)	Transformer	1 No. 9 Kg. DCP extinguisher per transformer.
(xxiii)	UPS / Charger Room	1 No. 2 Kg. CO ₂ extinguisher.

For pipeline installations, the portable extinguisher shall be provided as per the above list (4.6.1) suitably amended along with following additions:-

Sr. No.	Type of Area	Scale of Portable Fire Extinguishers
(i)	Main line pump shed (Engine/Motor Driven)	1 No. 75 Kg DCP, 10 Kg DCP & 6.8 Kg CO ₂ extinguishers per two pumps up to a maximum of 4 nos.
(ii)	Booster Pump	1 No. 10 Kg DCP per two pumps up to a maximum of 3 nos. and 1 No. 6.8 Kg CO ₂ extinguisher.
(iii)	Sump pump, Trans mix pump & oil water separator pump	1 No. 10 Kg DCP extinguisher.
(iv)	Scrapper Barrel	1 No. 10 Kg DCP extinguisher.
(v)	Control Room	2 Nos. 2.5 Kg Clean Agent and 1 No. 4.5 Kg CO ₂ extinguisher.
(vi)	UHF / Radio Room	2 Nos. 2.5 Kg Clean Agent and 1 No. 4.5 Kg CO ₂ extinguisher.
(vii)	Meter Filter Prover/Separator	1 No. 10 Kg DCP extinguisher.
(viii)	Repeater Station	1 No. 10 Kg DCP & 1 No. 2 Kg CO ₂ extinguisher.
(ix)	Mainline emergency equipment center	4 Nos. 10 Kg DCP & 2 Nos. 2 Kg CO ₂ extinguishers.
(x)	Air Compressor	1 No. 2 Kg CO ₂ & 1 No. 5 Kg DCP extinguisher.

4.7.2 WHEELED FIRE FIGHTING EQUIPMENT

For Installations having tanks of diameter larger than 9 m, following firefighting equipment shall be provided: -

Size of Terminal (In KL)	Water cum Foam Monitor (Number)



For installation having Aggregate above ground capacity upto 1000KL	Minimum 1 number of high-volume-long-range foam monitor having minimum capacity of 1000GPM.
For Installation having aggregate capacity above 1000KL & upto 1,00,000KL	Minimum 2 Numbers of high-volume long-range foam monitors having minimum capacity of 1500 GPM.
For Installation having aggregate capacity more than 1,00,000 KL	More than 2 Numbers of high-volume long-range foam monitors having minimum capacity of 1500 GPM

Foam compound trolley of minimum 1000 liters shall be provided along with compatible foam induction system for tanks having semi fixed foam pourer system.

4.7.3 HOSES, NOZZLES & ACCESSORIES

(i) HOSES

- Reinforced rubber lined non-percolating synthetic fire hoses conforming to IS- 636/ (Type 3) shall be provided.
- The length and diameter of the hoses shall be 15 m and 63 mm respectively fitted with instantaneous type male & female couplings of material as specified in IS 903.
- The number of spare hoses stored in an oil installation shall be 30% of the number of hydrant outlets. The minimum No. of hoses stored, however, shall not be less than 10.
- The hoses shall be stored at convenient and easily accessible location in the oil installation.

(ii) NOZZLES

In addition to the nozzle provided in each hose box, foam branch pipe and Water curtain nozzle as per relevant IS/UL Codes shall be maintained at the location.

(iii) ACCESSORIES

The following minimum no. of Personal Protective Equipment, First Aid Equipment & Safety Instrument shall be provided in emergency trolley as indicated against each item.

Stretcher with blanket: 2Nos.

First Aid box: 2Nos.

Rubber hand glove: 2 Pairs.

Explosimeter: 1 Nos.

Fire proximity suit: 2 Suit.

Resuscitator: 1 Nos.

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Electrical siren (3 Km range): 1 No.

Hand operated siren: One each at strategic locations such as Admn Bldg, Laboratory, T/L Loading/Unloading Facility, T/W Loading/Unloading Facility, Tank Farm, FW Pump House & Product Pump House (s).

Water jet blanket: 1 No.

Red & Green flag for fire drill: 2 Nos. in each colour.

SCBA Set (30-minute capacity): 2 set with spare cylinder. PA system:- 1 No.

Hose box: Between two hydrant points.

Fire hose: 2 Nos. per hose box.

Foam branch pipe: 2 Nos.

Multipurpose nozzle: 5 Nos.

The above guidelines are minimum requirement of each item and can be increased depending on the scale of operations/size of installation or requirement of Local Statutory Bodies/State Govt.

In addition, an emergency kit shall be provided consisting of safety items as per the attached list (**Annexure VII**) and shall be readily available at the terminals.

All the items of the kit shall be kept on a trolley specifically designed for the purpose. List of PPE's as mentioned in other clauses should be merged with this list.

4.7.4 MOBILE FIRE FIGHTING EQUIPMENT

In view of comprehensive Fixed and First Aid Fire protection equipment recommended in the standard, provision of Mobile firefighting equipment in the installation is not considered necessary. However, the requirement of such equipment should be reviewed keeping in mind the size, nature, and location of the installation.

Installation having aggregate storage capacity more than 5,00,000 kl should consider two variable flow mobile water cum foam monitors of minimum 4000 gpm capacity. In addition to this installation should have full-fledged manned fire station with minimum two multipurpose/foam fire tenders and trained fire crew having competency as per OISD-STD-228.

5.0 FIRE ALARM/COMMUNICATION SYSTEM

5.1 FIRE ALARM SYSTEM

- i) Hand operated sirens shall be provided at strategic locations and clearly marked in the installation.
- ii) Electric fire siren shall be installed at suitable location with operating switch located near the risk area at a safe, identifiable, and easily accessible place.

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- iii) Electric fire siren shall be audible to the farthest distance in the installation and also in the surrounding area up to 1 km from the periphery of the installation.
- iv) Electric fire sirens shall be connected to feeder to ensure continuous power supply during emergency shut down.
- v) ESD/Manual Call Point (MCP) shall be provided at various strategic locations as per company policy.
- vi) The following fire siren codes shall be followed for different emergency situations.
 - Emergency Level I – The siren code for Level I shall be decided by entity depending upon the size and complexity of installation. Absence of siren is not permitted.
 - Emergency Level II - A wailing siren for two minutes; and.
 - Emergency Level III- Same type of siren as in case of Level II but the same will be sounded for three times at the interval of one minute i.e. (waiting siren 2min + gap 1min + siren 2min +gap 1min +waiting siren 2min) total duration of disaster siren to be eight minutes.]
 - ALL CLEAR: Straight run siren for two minutes.
 - TEST: Straight run siren for two minutes at frequency at least once a week

5.2 COMMUNICATION SYSTEM

- i) Communication system like Telephone, Public Address System, etc. shall be provided in the installation.
- ii) In hazardous areas, flame-proof/intrinsically safe mobile or Paging System/ Walkie-talkie system or VHF/UHF Set shall be provided.

5.3 EMERGENCY POWER SUPPLY

Emergency lighting shall be provided for operating areas like the generator room, air compressor room, PCC/MCC room and control room. Emergency power supply shall also be provided to panels of all fire alarm/detection system/other firefighting system, critical valves linked with ESD, sectionalizing valves in mainline or any other equipment/room as decided as critical by operator. Emergency power supply in case of failure of regular power supply can be from DG/UPS/any other alternative reliable power supply. Emergency power supply shall be available as per process/equipment requirements, but within a period not exceeding 30 seconds from the instant of failure of normal supply. All critical equipment are to be identified and uninterrupted power backup to be provided. For critical lighting requirements, OISD-RP-149 shall be referred.

5.4 DETECTION AND ALARM SYSTEM

Hydrocarbon detectors shall be installed near all potential leak source of class-A e.g., tank dykes, tank manifolds, pump house manifold etc. Adequate flame or heat detection mechanism shall be provided for detection of ethanol fires as ethanol burns with a flame which is not clearly visible in bright sun light.

Hydrocarbon detector shall be maintained in line with OISD-STD153.

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6.0 FIRE PROTECTION SYSTEM AND EMERGENCY RESPONSE FOR CROSS COUNTRY PIPELINES

The pipeline system shall be equipped with an appropriate safety and fire protection system. The operating company shall have requisite plans and procedures in place for the operation and maintenance of the same including a detailed Emergency Response mechanism.

A risk assessment for each installation shall be carried out at design stage to determine the requirement of detection and alarm annunciation of hydrocarbon in atmosphere indicative of unintended release, fire / smoke in enclosed spaces and fire-fighting equipment to safeguard the installations from any fire hazard.

Notwithstanding the requirement arising from above risk assessment, minimum safety devices and fire-protection facilities as indicated in the following sections for different type of installations shall be implemented and maintained.

In addition to the requirements specified in the following sections for cross-country pipelines, the relevant provisions of this standard, along with applicable clauses from OISD-STD-141, OISD-STD-214, and OISD-STD-226, shall also be applicable.

6.1 FIRE WATER SYSTEM:

6.1.1 DESIGN BASIS:

The basis of design of fire protection facilities should presume that no external fire-fighting agencies would be available for Main Pump Station for a duration of minimum 4 hours.

The fire water pressure system shall be designed for a minimum residual pressure of 7.0 kg/cm² at the hydraulically remotest point.

In case of intermediate pigging stations of LPG & liquid pipelines, entity shall carry out risk assessment of pigging station. Based on risk assessment recommendations, entity should provide fire water networks (with medium velocity water sprinkler system for LPG). However, in case of Natural gas pipeline pigging station handling condensate, water network with medium velocity water sprinkler system shall be provided.

6.1.1(A) FOR CROSS COUNTRY LIQUID HYDROCARBON PIPELINE

Fire water system shall be designed based on single largest fire contingency or two largest fire contingencies in the installation as per clause No 4.2.2. Fire water flow for pump house shed for pipeline installations shall be at a rate of 10.2 lpm/m².

Foam requirement shall be calculated as follows:

- I. Mainline pump shed shall be covered by minimum two water cum foam monitors from ~~four~~ different directions. The design of pump shed sheeting shall be such that throw of these monitors can reach at target hazard (i.e. pump/engine/motor/suction-discharge MOV).
- II. In case of obstruction due to pump shed sheeting/ structural, additional provision of firefighting with foam mixture shall be provided as per below:
 - a. Automatic foam spray system shall be provided in the pump shed. Foam tank of this system along with all associated actuating valves, shall be at a safe distance of minimum 15 m from the pump shed. A fire wall shall be provided for the protection of

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the deluge valve and for operating personnel.

- b. Foam application rate in this case shall be 4.1 lpm/m². The foam spray system including nozzles and piping shall be designed accordingly.
- III. A retaining wall of 230 mm width, 300 mm height around pump shed flooring edge shall be provided, to retain foam solution blanket inside pump shed.
- IV. Foam requirement shall be calculated for a minimum period of 30 minutes for applicable scenario listed above at s.no. I or II.

6.1.1(B) FOR LPG CROSS COUNTRY PIPELINE

Fire Water System shall be designed to address single largest risk in the pumping station as a minimum.

The fire water system in the plant shall be designed to meet the highest fire water flow requirement of a single largest area risk at a time plus 288 m³/hr for operating 2 Nos. of fire water monitors/supplementary hose requirements.

Pump House Area shall be considered as single risk area. Alternatively, it can be divided into suitable number of zones with minimum 10 m width.

Where Despatch / Receiving Terminals are co-located in any marketing location or refinery or any plant location, fire water network with hydrants and monitors in the network of such installation, if available, shall be extended.

The Fire water pumping requirement for medium velocity spray system having auto operated deluge valve shall be calculated based on following cooling rate:

Pump Shed: 20.4 lpm/m² area of the pump shed to be calculated considering outer foundation column distances.

Scraper area, Metering area, Filtering area and receipt/delivery manifold area: 10.2 lpm/m² of surface area to be considered.

6.1.1(C) FOR NATURAL GAS PIPELINES:

Fire Water System shall be designed to address single largest risk in the Compressor station as a minimum.

All Compressor Area, Pig Launcher and Receivers (Scraper Area), Metering Area, Filtering Area, Condensate Collection Area, or any other critical area. shall be covered by medium velocity spray system. For other areas, Operator should carry assessment to establish the requirement of medium velocity spray system.

Compressor Area shall be considered as single risk area. Alternatively, it can be divided into suitable number of zones with minimum 10 m width.

The Fire water pumping requirement for medium velocity spray system having auto operated deluge valve shall be calculated based on following cooling rate:

Compressor area: 20.4 litres per min per square meter area (lpm/m²) of the compressor area.

Scraper Area and Filtering Area: 10.2 lpm/m² of process area to be considered.

Other Area: 3 lpm/m² of process area to be considered.

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Where Despatch / Receiving Terminals are co-located in any consumer premises having fire water network with hydrants and monitors in the network of such consumer installation if available, shall be extended.

6.2 FIRE WATER STORAGE: Refer Cl.4.3.3

6.3 FIRE WATER PUMPS: Refer Cl.4.3.4

6.4 FIRE HYDRANT NETWORK

Fire hydrant network shall be in closed loops to ensure multidirectional flow in the system. Isolation valve shall be located near loop junction. Additional Isolation valves shall be provided where the length of the pipe section is more than 300 m.

Fire water ring main shall be sized for 120% of the design water flow rate. Velocity of the water shall not exceed more than 5 m/s in the fire water ring main.

In case of saline or brackish water or treated effluent water is used, the fire water main of steel pipes shall be internally cementing mortar lined, or glass reinforced epoxy coated or made of pipe material suitable for the quality of water. Alternatively, pipes made of composite materials shall be used.

A fire water ring main shall be provided all around perimeter of the plant facilities. At least one hydrant post shall be provided for every 30 m of external wall measurement or perimeter of the battery limit. In addition to Hydrant posts, Monitors shall be installed in such a way that each facility is covered by minimum two monitors from two different directions. Owner may consider providing more water cum foam monitors / water monitors in place of fire hydrant/hoses considering limited manpower for firefighting during odd hours, especially at smaller pump stations.

Additionally, one portable water cum foam monitor (trolley mounted) of 750/900 GPM per installation should be considered, which can be used at any location as per requirement by connecting fire hose.

Fire water network shall be provided with pressure transmitter at hydraulically remotest point for monitoring network pressure from the control room.

Fire water ring main shall be laid above ground at a height of 300 mm to 400 mm above finished ground level. Pipes made of composite material shall be laid underground.

The mains shall be supported at regular intervals not exceeding 6 m. For pipeline size less than 150 mm diameter, support interval shall not be more than 3 m.

The ring main may be laid underground at the following places:

(i) Road crossings

(ii) Where above ground piping is likely to cause obstruction to operation, vehicle movement and likely to get mechanical damage.

The main shall have 1 m earth cover in open ground and 1.5 m cover under the roads.

The underground fire water network shall be provided with suitable coating /wrapping or

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concrete / masonry.

Double headed hydrants with two separate landing valves on 4" stand post shall be used at 1.2 m above working platform level/ground level. Individual isolation valve shall be provided for each hydrant post.

Fire water monitors shall be provided with independent isolation valves at 1.2 m above working platform level/ground level. Hose Box with 2 Nos. of hoses, valve key and a nozzle shall be provided at each hydrant point.

Fire Hydrants/ monitors shall be located at a minimum distance of 15 m from the hazardous facility / equipment. In case of buildings, this distance shall not be less than 2 m and not more than 15 m from the face of building. Provisions of hydrants within the building shall be provided in accordance with IS: 3844.

In case of liquid hydrocarbon pipeline, AFFF foam shall be provided at each water cum foam monitor in suitable size tank/drums like 200 liters so as to avoid repeated changeover of small size cans.

6.4.1 MEDIUM VELOCITY WATER SPRAY SYSTEM (MVWS) FOR CROSS COUNTRY LPG PIPELINES & CROSS-COUNTRY NATURAL GAS PIPELINES:

In case of LPG pipelines, all LPG pumps (booster and mainline pumps), Pig launcher and receivers, metering area, filtering area and receipt & delivery manifold or any other critical area with flange connections area shall be fully covered by medium velocity spray system connected with fire water network.

In case of natural gas pipelines, All Compressor Area, Pig Launcher and Receivers (Scrapper Area), Metering Area, Filtering Area, Condensate Collection Area etc. shall be covered by medium velocity spray system for other areas.

Heat detection through thermal fuses / quartz bulbs (QB to blow at 79 °C) / Electro pneumatic (EP) detectors) / flame detector for detection of fire for automatic actuation of medium velocity water spray system shall be provided. The QB / EP detectors shall be placed at 300 mm from hazards. In areas without specific hazard, detectors shall be placed evenly across the ceiling or with a maximum spacing of 3 m inside the shed.

Provision should be made for actuation of auto operated MVWS through Gas Monitoring System i.e., Hydrocarbon Detectors preferably point type.

The deluge valve should be located at 15 m from the risk being protected. A fire wall shall be provided for the protection of the deluge valve and for operating personnel.

Provision should be made to operate deluge valves from control room also to facilitate additional protection in case of deluge valve fails to operate automatically.

The medium velocity spray system provided at all critical areas shall have spray nozzles directed radially to the facilities intended for cooling preferably at a distance of 0.6 m from the surface of the equipment / facility. Care shall be taken in positioning nozzles so that water spray does not miss the targeted surface and reduce the efficiency or calculated discharge rate.

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All spray nozzles shall be inspected for proper positioning, corrosion and cleaned, if necessary, at an interval of not more than 12 months or earlier based on the experience.

6.5 HOSES, NOZZLES, ACCESSORIES AND SPARES

The following spare hoses, nozzles, accessories and spares should be maintained:

Fire hoses – 2 nos./per hose box/per Hydrant Point or Minimum 10 Nos.

In addition to the nozzles provided in the hose boxes there shall be 2. 25% spare hoses shall be stored. 20% of total extinguishers (considering each type) shall be maintained for replacements at designated stores.

For cross country Liquid hydrocarbon pipeline:

First aid firefighting equipment like portable fire extinguishers, Wheeled Fire Fighting Equipment, Foam compound trolley, Hoses, Nozzles, Accessories etc. shall be provided at pipeline installations as per Cl.4.7.

6.5.1 FIRST AID FIRE FIGHTING EQUIPMENT FOR NATURAL GAS & LPG CROSS COUNTRY PIPELINE

Pump / Compressor Station		
SN	Area	Specifications
1.	LPG Pump House / Compressor area	1 x 9 kg DCP / 50 m ² or Minimum 2 Nos. whichever is higher, 1 x 75 kg DCP / 750 m ² or Minimum 1 No. whichever is higher and 1 x 4.5 / 6.5 kg CO ₂ or Clean Agent / 250 m ² or Minimum 2 No. whichever is higher.
2.	Other areas (LPG)	1 x 9 kg DCP / 250 m ² or Minimum 2 Nos. whichever is higher and having maximum travelling distance of 15 m, 1 x 75 kg DCP / 750 m ² or Minimum 1 No. whichever is higher and 1 x 4.5 kg / 6.5 kg CO ₂ / 250 m ² .
3.	Other areas (Natural Gas)	1 x 9 kg DCP / 250 m ² or Minimum 2 Nos. whichever is higher and having maximum travelling distance of 15 m, 1 x 75 kg DCP / 750 m ² or Minimum 1 No. whichever is higher. Ensure minimum 1 x 9 kg DCP at Scraper, Filtering and Metering area and 1 x 4.5 / 6.5 CO ₂ / 250 m ² .
4.	Office / Canteen / Stores	1 x 9 kg DCP / 250 m ² or Minimum 2 Nos. whichever is higher and having maximum travelling distance of 15 m, 1 x 4.5 / 6.5 kg CO ₂ or Clean Agent / 250 m ² or Minimum 2 Nos. whichever is higher
5.	Control Room / MCC / DG Room / HT room	2 x 4.5 / 6.5 kg CO ₂ / Clean Agent in each room or per 100 m ² floor area.
Dispatch / Receiving Terminal		
6.	Terminal Area	1 x 9 kg DCP / 250 m ² or Minimum 2 Nos. whichever is higher and having maximum travelling distance of 15 m, 1 x 75 kg DCP / 750 m ² or Minimum 1 No. whichever is higher.
7.	Office / Canteen /	1 x 9 kg DCP / 250 m ² or Minimum 2 Nos. whichever is



	Stores	higher and having maximum travelling distance of 15 m, 1 x 4.5 / 6.5 kg CO ₂ or Clean Agent / 250 m ² or Minimum 2 Nos. whichever is higher
8.	Control Room / MCC / DG Room / HT room	2 x 4.5 / 6.5 kg CO ₂ / Clean Agent in each room or per 100 m ² floor area.
Intermediate pigging station		
9.	Pigging area	1 x 9 kg DCP / 250 m ² or Minimum 2 Nos. whichever is higher and having maximum travelling distance of 15 m, 1 x 50 / 75 kg DCP / 750 m ² or Minimum 1 No. whichever is higher.
10.	Electrical / Battery / Panel Room	1 x 4.5/6.5 kg CO ₂ or Clean Agent in each room or Minimum 2 Nos. whichever is higher.
Sectionalizing Valve Station		
11.	Sectionalizing Valve Area	1 x 9 kg DCP / 250 m ² or Minimum 2 Nos. whichever is higher and having maximum travelling distance of 15 m.
12.	Electrical / Battery / Panel Room	1 x 4.5 / 6.5 kg CO ₂ or Clean Agent in each room or Minimum 2 Nos. whichever is higher.

6.6 MATERIAL SPECIFICATIONS

Refer clause 4.3.7.

6.7 INSPECTION

Inspection of firefighting equipment and system shall be carried out in line with OISD-STD-142.

6.8 FIRST AID AND SAFETY ACCESSORIES

Windsock shall be provided at Pump/ Compressor Station, Dispatch terminal, Receipt terminal, Intermediate Pigging Station & Sectionalizing Valve Station on an appropriately elevated structure like the control room, building, fire water pump house etc. in such a manner so as to avoid blind areas.

6.8.1 LPG CROSS COUNTRY PIPELINE

A trolley or facility containing first aid and safety accessories shall also be provided at Pump Station, Receiving Terminal, Intermediate Pigging Station and Sectionalizing Valve Station. The minimum accessories / first aid items not limited to following shall be provided:

a) AT PUMP STATION AND DESPATCH / RECEIVING TERMINAL

- (i) Safety helmets-1 No./Person (min 10 nos.).
- (ii) Stretcher with blanket- 2 Nos.
- (iii) First aid box - 2 Nos.
- (iv) Rubber hand gloves for electrical purpose - 2 Nos.
- (v) Low temperature rubber hand gloves - 4 Pairs.
- (vi) Low temperature protective clothing - 2 Sets.

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- (vii) PVC/Chemical protective clothing – 2 Sets
- (viii) Fire proximity suit -2 No.
- (ix) Resuscitator – 2 Nos.
- (x) Self-contained breathing apparatus with one spare cylinder (Capacity 30 min) –2 No.
- (xi) Water jel blanket- 2 Nos.
- (xii) Portable Multi Sensor Gas Detector - 2 Nos.
- (xiii) Megaphone (EX-PROOF) – 1 No.
- (xiv) Safety Torch – 2 No.

b) AT INTERMEDIATE PIGGING STATION AND SECTIONALIZING VALVE STATION

- (i) Safety helmets - min 02 Nos.
- (ii) First aid box - 1 No.
- (iii) Rubber hand gloves for electrical purpose - 1 No.
- (iv) Low temperature rubber hand gloves - 2 Pairs
- (v) PVC/Chemical protective clothing – 1 No.
- (vi) Resuscitator – 1 No.
- (vii) Water jel blanket- 1 no.
- (viii) Portable Oxygen cum Flammable detectors– 1 nos.
- (ix) Safety Torch – 1 No.

Installations, if having provision of Emergency Response Vehicle (ERV) equipped with above mentioned equipment are excluded from providing such trolley at site.

Windsock shall be provided at Pump Station, Receiving Terminal, Intermediate Pigging Station & Sectionalizing Valve Station on an appropriately elevated structure like the control room, building, fire water pump house etc. in such a manner to avoid blind areas.

6.8.2 NATURAL GAS PIPELINES:

A trolley or facility containing first aid and safety accessories shall also be provided at Compressor Station, Dispatch terminal, Receipt terminal, Intermediate Pigging Station and Sectionalizing Valve Station.

6.9 FIRE DETECTION AND ALARM SYSTEM

For Cross Country Liquid Hydrocarbon Pipeline Clause No. 13.3 of OISD-STD-141 shall be applicable.

For Cross Country LPG Pipelines Clause No.10.7, 10.8 & 10.9 of OISD-STD-214 shall be applicable.

For Cross Country Natural Gas Pipelines Clause No.10.7 & 10.8 of OISD-STD-226 shall be applicable.

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6.10 COMMUNICATION SYSTEM

For Cross Country Liquid Hydrocarbon Pipeline Clause No. 13.6 of OISD-STD-141 shall be applicable.

For Cross Country LPG Pipelines Clause No.10.11 of OISD-STD-214 shall be applicable.

For Cross Country Natural Gas Pipelines Clause No.10.9 of OISD-STD-226 shall be applicable.

6.11 EMERGENCY SHUTDOWN SYSTEM

Emergency shutdown system for cross country pipelines shall be provided as per their respective standards i.e., OISD-STD-141, OISD-STD-214, OISD-STD-226.

6.12 EMERGENCY RESPONSE & DISASTER MANAGEMENT PLAN

An Emergency response and Disaster Management Plan for handling onsite and offsite emergencies due to oil leak, burst or fire shall be prepared in accordance with applicable statutory/ regulatory/ OISD guidelines such as OISD-GDN-227/ MOP&NG DMP/ Petroleum and Natural Gas Regulatory Board (Codes of Practices for Emergency Response and Disaster Management Plan (ERDMP) Regulations, 2010 and amendments thereof. The ERDMP documents shall be prepared taking into consideration the HAZOP & QRA studies and critical emergency scenarios entailed in it. The hazards shall be evaluated under Level 1, Level 2 and Level 3 type with its consequences and operational controls.

A mutual aid scheme for the fire fighting and emergency rescue operations shall be in place involving nearby industries. Such mutual aid scheme shall be activated during emergency drills (Onsite and Offsite), and the response of the mutual aid partners shall be recorded and maintained. Frequency of Onsite emergency drill and Offsite emergency drill shall be as per approved ERDMP. Each station shall carry out at least one offsite (level III) mock drill once a year in the pipeline stretch under its jurisdiction. The findings / short comings of such drills shall be recorded, analysed and corrective actions shall be taken.

7.0 BATTERY FIRE

From consumer electronics and electric vehicles to industrial equipment and energy storage systems, lithium-ion batteries can be found everywhere. Due to the materials utilized to manufacture these batteries, the scene of a lithium-ion battery fire can best be compared to a hazmat (hazardous material) incident. Although extinguishing fire is a primary concern, flammability, explosivity and toxicity must be addressed to ensure proper protection and prevent re-ignition.

In a cell or battery, ignition occurs when lithium-ion batteries enter a chemical process called thermal runaway. Cells produce heat and emit explosive and toxic gases, developing a self-propelling loop that leads to thermal runaway propagation. This chain reaction will continue to impact adjacent cells until thermal runaway propagation is halted. Confined spaces such as server rooms, automobiles, aircraft, spacecrafts, shipping containers, energy storage systems and more can pose an increased risk.

During the battery fire, a flammable electrolyte used within cells catches fire. Thus, fire starts inside a cell which makes it difficult to reach. All the conventional firefighting media, including

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plain water, works at the macro / surface level to remove heat. In the case of battery fire, due to deep seated nature of fire, the effectiveness of conventional firefighting media is very less and thus they are required in humongous quantity to achieve extinguishment.

In many battery fires, one of the byproducts of the oxidation (burning) chemical process is oxygen, thus the fire is fed by creating its own oxygen. Hence, any firefighting media / techniques that works on a principle of oxygen depletion find limited success.

7.1 FIRE PROTECTION

For fire protection of Li ion battery, Encapsulator Agent (EA) based on spherical micelle technology should be used in line with NFPA 18A & NFPA 855 (latest edition). (For details refer Annexure-VIII)

8.0 EMERGENCY RESPONSE/ & TRAINING

8.1 ORGANOGRAM

A well-defined comprehensive Emergency response organogram shall be developed in line with OISD-STD-227.

8.2 TRAINING

- i) All the plant personnel shall be trained on fire prevention and firefighting aspects in line with OISD-STD-154. The safety, rescue operation and firefighting training shall be compulsory for all officers, operators, security, T/T drivers & contract workmen, clericals who are likely to be present in the installation & record maintained.
- ii) Every employee or authorized person of contractor working in the installation shall be familiarized with fire siren codes and the location of fire siren operating switch nearest to his place of work.
- iii) Instructions on the action to be taken in the event of fire should be pasted at each siren point and familiarity with these instructions ensured and recorded.
- iv) Monthly fire drills considering various scenarios shall be conducted regularly with full involvement of all employees of the installation. The mock drill shall include the full shut down system activation once in six months.
- v) The offsite disaster mock drills shall be conducted periodically as per local statutory requirements. The company should approach and coordinate with the district authority for conducting "Offsite Mock Drills".
- vi) The post drill analysis should be carried out & discussed emphasizing areas of improvements.
- vii) The record of such drills should be maintained at the location.
- viii) Security staff should be trained for firefighting and rescue operation.

8.3 MUTUAL AID

Installation should have written mutual aid agreements with similar neighbouring industries fully detailing the responsibilities of the members of the scheme, the procedures to be adopted, the minimum number of equipment and manpower and minimum quantity of

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consumables to be exchanged/loaned in line with OISD-STD-227.

8.4 FIRE EMERGENCY MANUAL

- 8.4.1 Each installation shall prepare a Comprehensive fire emergency manual covering all emergency scenarios outlining the actions to be taken by each personnel in the event of fire emergency effective handling and the same shall be available to all personnel in the installation.
- 8.4.2 The key action points of this manual shall be displayed at strategic locations in the installation for ready reference.

9.0 FIRE PROTECTION SYSTEM, INSPECTION AND TESTING

- i) The fire protection equipment shall be kept in good working condition all the time.
- ii) The fire protection system shall be periodically tested for proper functioning and logged for record and corrective actions.
- iii) One officer shall be dedicated and made responsible for inspection, maintenance & testing of fire protection system.
- iv) The responsibilities of each officer shall be clearly defined, explained, and communicated to all concerned in writing for role clarity.
- v) The requirements of OISD-STD-142 in respect of periodic inspection, maintenance & testing of firefighting equipment shall be complied with.

9.1 FIRE WATER PUMPS

- i) Fire water pumps should be tested in line with OISD-STD-142.
- ii) The jockey pump shall be tested weekly for auto start and stop.

9.2 FIRE WATER RING MAINS

- (i) The ring main shall be checked for leaks once in a year by operating one or more pumps & keeping the hydrant points closed to get the maximum pressure.
- (ii) The ring mains, hydrant, monitor & water spray header valves shall be visually inspected for any missing accessories, defects, damage, and corrosion every month and records maintained.
- (iii) All valves on the ring mains, hydrants, monitors & water spray headers shall be checked for leaks, smooth operation and lubricated once in a month.

9.3 FIRE WATER SPRAY SYSTEM

- i) Water spray system shall be tested for performance i.e., its effectiveness & coverage once in six months.
- ii) Spray nozzles shall be inspected for proper orientation, corrosion and cleaned, if necessary, at least once a year.
- iii) The strainers provided in the water spray system shall be cleaned once in a quarter and records maintained.

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9.4 FIXED/SEMI FIXED FOAM SYSTEM

Fixed/Semi fixed foam system on storage tanks shall be tested in line with OISD-STD-142. The foam maker/chamber should be designed suitably to facilitate discharge of foam outside the cone roof tank. After testing foam system, piping should be flushed with water.

9.5 CLEAN AGENT SYSTEM

Clean agent fire extinguishing system should be in line with OISD-STD-142.

9.6 HOSES

Fire hoses shall be tested in line with OISD-STD-142.

9.7 COMMUNICATION SYSTEM

Electric and hand operated fire sirens should be tested for their maximum audible range once a week.

9.8 FIRE WATER TANK/RESERVOIR

- i) Above ground fire water tanks should be inspected externally & internally as per OISD-STD-129.
- ii) The water reservoir shall be inspected periodically not more than 3 year to ensure the integrity of the reservoir and there is no seepage and leakage from the reservoir. During such inspection, any accumulation of mud, sand, weeds, and other undergrowth, which can reduce the capacity of the reservoir and obstruct free flow of water, shall be removed and record to be maintained. The required remedial action shall be taken.

9.9 FIRE EXTINGUISHERS

Inspection, testing frequency and procedure should be in line with OISD-STD-142.

10.0 REFERENCES

1. NFPA 11(2024): Standard for Low Expansion Foam Systems
2. NFPA 12(2025): Standard on Carbon Dioxide Extinguishing Systems.
3. NFPA 13(2025): Standard for The Installation of Sprinkler Systems
4. NFPA 14(2024): Standard for the Installation of Standpipe and Hose Systems
5. NFPA 15(2022): Standard for Installation of Water Spray Systems
6. NFPA 17(2024): Standard for Dry Chemical Extinguishing Systems
7. NFPA 18A(2022): Standard on Water Additives for Fire Control and Vapor Mitigation
8. NFPA 20(2025): Standard for Installation of Centrifugal Fire Pumps
9. NFPA 24: Standard for the Installation of Private Fire Service Mains and Their Appurtenances
10. NFPA 30(2024): Flammable and Combustible Liquids
11. NFPA 72: National Fire Alarm and Signalling Code

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12. NFPA-2001: Standard for Clean Fire Extinguishing System
13. NFPA-855: Standard for the Installation of Stationary Energy Storage Systems
14. Petroleum Rules -2002
15. International Safe Practices of Oil Industry
16. OISD-GDN-115: Guidelines on specification of Fire Fighting Equipment & Mitigation Strategy for Credible Fire Scenario
17. OISD-STD-118(2025): Layouts for Oil and Gas Installations
18. OISD-STD-141: Cross Country Liquid Hydrocarbon Pipeline
19. OISD-STD-142: Inspection of Fire Fighting Equipment & Systems
20. OISD-STD-154: Safety Aspects in Functional Training
21. OISD-STD-214: Cross Country LPG Pipelines
22. OISD-STD-226: Cross Country Natural Gas Pipelines & City Gas Distribution Network
23. OISD-STD-227: Emergency Response and Preparedness in E&P Industries
24. Oil Mines Regulations, 2017
25. Ozone Depletion Substances Regulation & Control Rules 2000, Ministry of Environment & Forests, Government of India
26. IS-15683: Portable Fire Extinguishers - Performance and Construction - Specification
27. IS:14609 : Dry Chemical Powder for fighting A,B,C Class Fire – Specification
28. IS 5: Colours for ready mixed paints and enamels
29. IS 15683: Portable Fire Extinguishers Performance and Construction Specification
30. IS 26: Selection, installation, and maintenance of first-aid fire extinguishers – Code of practice
31. UL154: Carbon-Dioxide Fire Extinguishers
32. UL299: Dry Chemical Fire Extinguishers
33. UL711: Rating & Fire Testing of Fire Extinguishers
34. UL162: Foam Equipment and Liquid Concentrates
35. T4S : Technical Standards and Specifications including Safety Standards for Petroleum installation

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11.0 ABBREVIATION:

AFFF: Aqueous Film Forming Foam

AR-AFFF: Alcohol Resistant Aqueous Film Forming Foam

FFFP: Film Forming Fluoro-Protein

HVLRM: High-volume long-range monitor

12.0 CONVERSION TABLE:

1 L/min 0.06 m³/hr

1 gpm 0.228 m³/hr

DRAFT

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ANNEXURE- I

SAMPLE CALCULATION OF FIRE WATER FLOW RATE ^{Note1} (Clause 4.3.2 to be applicable)

1. DESIGN BASIS

The fire water system in an installation shall be designed to meet:

- The fire water flow requirement of fighting single largest fire contingencies for locations where total aggregated storage capacity in the location is upto 30,000KL.
- The fire water flow requirement of fighting two largest fire contingencies simultaneously for all locations where total aggregated storage capacity in the terminal is more than 30,000 KL.

2. FIRE WATER DEMAND FOR SINGLE LARGEST FIRE (For locations with aggregate storage capacity upto 30000KL)

Consider various areas under fire and calculate fire water demand for each area based on design basis as indicated below, however, actual tank dimensions available in the terminal shall be considered.

2.1 FIRE WATER FLOW RATE FOR FLOATING ROOF TANK PROTECTION

Data

Total storage capacity in one dyke area	=	20,000 m ³ .
No. of tanks	=	2.
Capacity of each tank	=	10,000 m ³ .
Diameter of each tank	=	30 m.
Height of each tank	=	14.4 m.

a) Cooling water flow rate

(i) Cooling water required for tank on fire

Cooling water rate	=	3 lpm/m ² of tank area for tank on fire.
Cooling water required	=	$\pi \times 30 \text{ m} \times 14.4 \text{ m} \times 3 \text{ lpm/m}^2$.
	=	4069.4 lpm.
	=	$\frac{4069.4 \times 60}{1000} \text{ m}^3/\text{hr} = 244 \text{ m}^3/\text{hr}.$

Assuming that second tank is also located within the same tank. Therefore, in such case cooling required is at the rate of 3 lpm/m² of tank shell area.

Note-1: These are sample calculations only. Calculations on the basis of actual site conditions and dimensions need to be carried out for each installation as per guidelines provided in clause 4.3.

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(ii) Cooling water required for tanks in same dyke area

$$\begin{aligned}
 \text{Cooling water rate} &= 3 \text{ lpm/m}^2 \text{ of tank area.} \\
 \text{Cooling water required} &= \pi \times 30 \text{ m} \times 14.4 \text{ m} \times 3 \text{ lpm/m}^2. \\
 &= 4069.4 \text{ lpm.} \\
 &= \frac{4069.4 \times 60 \text{ m}^3/\text{hr}}{1000} \\
 &= 244 \text{ m}^3/\text{hr}
 \end{aligned}$$

Total Water required for cooling of tanks (item i + ii) = 244 + 244 = 488 m³/hr

b) Foam water flow rate

Water flow required for applying foam on a largest tank burning surface area (rim seal area)

For floating roof tank of 30 M diameter,

$$\begin{aligned}
 \text{Diameter of the tank (D1)} &= 30 \text{ M} \\
 \text{Distance of foam dam from shell} &= 0.8 \text{ M} \\
 \text{Diameter of roof up to foam dam (D2)} &= 30 - (2 \times 0.8) = 28.4 \\
 \text{Rim seal area} &= (\pi/4) \times (30^2 - 28.4^2) \\
 &= (\pi/4) \times 93.44 \\
 &= 73.4 \text{ m}^2 \\
 \text{Foam solution rate @ 12 lpm/m}^2 &= 880.8 \text{ lpm} \\
 \text{Foam water required} &= 0.97 \times 880.8 \text{ lpm} \\
 \text{(For 3\% foam concentrate)} &= 854.4 \text{ lpm.} \\
 &= \frac{854.4 \times 60 \text{ m}^3/\text{hr}}{1000} \\
 &= 51 \text{ m}^3/\text{hr.}
 \end{aligned}$$

Total water flow rate (item a + item b) for floating roof tank protection:

$$\begin{aligned}
 \text{(i) Tank cooling} &= 488 \text{ m}^3/\text{hr.} \\
 \text{(ii) Foam solution application} &= 51 \text{ m}^3/\text{hr.} \\
 \text{Total (item i + ii)} &= 539 \text{ m}^3/\text{hr.}
 \end{aligned}$$

2.2 FIRE WATER FLOW RATE FOR CONE ROOF TANK PROTECTION

Data

$$\begin{aligned}
 \text{Total storage capacity in one dyke area} &= 10,000 \text{ m}^3. \\
 \text{No. of tanks} &= 2 \\
 \text{Capacity of each tank} &= 5000 \text{ m}^3. \\
 \text{Diameter of each tank} &= 24 \text{ m.} \\
 \text{Height of each tank} &= 12 \text{ m.}
 \end{aligned}$$

a) Cooling water flow rate



(i) Cooling water required for tank on fire

$$\begin{aligned}\text{Cooling water rate} &= 3 \text{ lpm/m}^2 \text{ of tank area for tank on fire.} \\ \text{Cooling water required} &= \pi \times 24 \text{ m} \times 12 \text{ m} \times 3 \text{ lpm/m}^2 \\ &= 2715.4 \text{ lpm.} \\ &= \frac{2715.4 \times 60}{1000} \text{ m}^3/\text{hr} \\ &= 163 \text{ m}^3/\text{hr}\end{aligned}$$

Assuming that other tank is also located within the same tank dyke. Therefore, in such case cooling required is at the rate of 3 lpm/m² of tank shell area.

(ii) Cooling water required for tank falling within same dyke area

$$\begin{aligned}\text{Cooling water rate} &= 3 \text{ lpm/m}^2 \text{ of tank area.} \\ \text{Cooling water required} &= \pi \times 24 \text{ m} \times 12 \text{ m} \times 3 \text{ lpm/m}^2 \\ &\quad + \pi/4 \times 24 \text{ m} \times 24 \text{ m} \times 3 \text{ lpm/m}^2 \\ &= 3089.06 \text{ lpm.} \\ &= \frac{3089.06 \times 60}{1000} \text{ m}^3/\text{hr} \\ &= 185 \text{ m}^3/\text{hr}\end{aligned}$$

$$\text{Total cooling water required (item i + ii)} = 163 + 185 = 348 \text{ m}^3/\text{hr.}$$

b) Foam water flow rate

$$\begin{aligned}\text{Foam solution application rate} &= 5 \text{ lpm/m}^2 \text{ of liquid surface area.} \\ \text{Foam solution required} &= \frac{\pi \times (24 \text{ m})^2 \times 5 \text{ lpm/m}^2}{4} \\ &= 2262.9 \text{ lpm.} \\ \text{Foam water required} &= 0.97 \times 2262.9 \text{ lpm} = 2195 \text{ lpm.} \\ \text{(For 3\% foam concentrate)} &= \frac{2195 \times 60}{1000} \text{ m}^3/\text{hr} \\ &= 132 \text{ m}^3/\text{hr.}\end{aligned}$$

Total water flow rate (item a + b) for cone roof tank protection:

$$\begin{aligned}\text{(a) Tank cooling} &= 348 \text{ m}^3/\text{hr.} \\ \text{(b) Foam solution application} &= 132 \text{ m}^3/\text{hr.} \\ \text{Total} &= 480 \text{ m}^3/\text{hr.}\end{aligned}$$

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2.3 FIRE WATER FLOW RATE FOR COOLING POL TANK WAGON LOADING GANTRY

a) Data

Total No. of loading points = Conventional or BTPN.
Width of tank wagon gantry = 12 m.
(Cooling two spur)

b) Cooling water flow rate

Divide total area of gantry into equal segments such that each segment measuring 15 m X 12 m and consider 3 segments operating at a time.

Water rate required = $3 \times 15 \text{ m} \times 12 \text{ m} \times 10.2 \text{ lpm/m}^2$.
= 5508 lpm
= **330 m³/hr**

2.4 Fire water calculation for full surface fire on largest floating roof tank (roof sinking case)

Data:

Total storage capacity in one dyke area = 20000 m³
No. of tanks = 2
Capacity of each tank = 10,000 m³
Diameter of each tank = 30 m
Height of each tank = 14.4 m

a) Cooling water requirement:

Cooling water rate @ 3 lpm/ m² of tank shell area for tank-on-fire
Cooling water required = $\pi \times 30 \times 14.4 \times 3$
= 4073.1 lpm
= **244 m³/hr**

Assuming that second tank is located within the tank dyke at a distance more than 30M from the tank shell.

Then, cooling water requirement @ 3 lpm/ m² of tank shell area = $\pi \times 30 \times 14.4 \times 3$
= 4069.4 lpm
= **244 m³/hr.**

Total cooling water = $(244 + 244) \text{ m}^3/\text{hr}$
= **488 m³/hr**

b) Water requirement in foam application

Foam Application Rate @ 9.75 lpm/m²

Foam Solution Requirement = $(\pi \times 30 \text{ m} \times 30 \text{ m}) / 4 \times 9.75 \text{ lpm/m}^2$
= 6894.6 lpm
= **414 m³/hr**

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$$\begin{aligned}\text{Water required for the foam solution} &= 0.97 \times 344\,414 \text{ m}^3/\text{hr} \\ &= 402 \text{ m}^3/\text{hr}\end{aligned}$$

Total water required for roof sink case:

Tank cooling	488 m ³ /hr
Foam application	402 m ³ /hr
Total	890 m³/hr
Total water requirement = 890 m³/hr	

Note: Full surface fire of floating roof tank roof sinking case being a remote possibility, it is considered as a single largest contingency for the purpose of arriving at design fire water requirement.

2.5 TOTAL DESIGN FIRE WATER FLOW RATE FOR SINGLE FIRE CONTEGENCY

The total fire water flow requirement will be highest of one of the fire water requirement calculated in 2.1 (539m³/hr) & 2.2 (480m³/hr), 2.3 (330 m³/hr) and 2.4 (890 m³/hr) above i.e., 890 m³/hr plus supplementary water (36 x 4 = 144 m³/hr) = 1034m³/hr.

Note:- The fire water demand shall be determined considering 65 minutes (additional 30minutes if foam stored for extended period) of foam application and 4 hours of tank cooling (including tank on fire) in the same dyke enclosure with supplementary water requirements for four hours.

**3.0 FIRE WATER DEMAND FOR TWO MAJOR FIRES SIMULTANEOUSLY
(For locations with aggregate storage capacity more than 30000KL &
less than 100000 KL)**

Consider various areas under fire and calculate fire water demand for each area based on design basis as indicated below, however, actual tank dimensions available in the terminal shall be considered.

3.1 FIRE WATER FLOW RATE FOR FLOATING ROOF TANK PROTECTION**Data**

Total storage capacity in one dyke area	=	32,000 m ³ .
No. of tanks	=	2.
Capacity of each tank	=	16,000 m ³ .
Diameter of each tank	=	40 m.
Height of each tank	=	14.4 m.

a) Cooling water flow rate**(i) Cooling water required for tank on fire**

$$\begin{aligned}\text{Cooling water rate} &= 3 \text{ lpm/m}^2 \text{ of tank area for tank on fire.} \\ \text{Cooling water required} &= \pi \times 40 \text{ m} \times 14.4 \text{ m} \times 3 \text{ lpm/m}^2 \\ &= 5430.8 \text{ lpm.} \\ &= \frac{5430.8 \times 60}{1000} \text{ m}^3/\text{hr} \\ &= 326 \text{ m}^3/\text{hr}\end{aligned}$$

Assuming that second tank is also located within the same tank dyke at a distance more than 30 m from the tanks shell. Therefore, in such case cooling required is at the rate of 1 lpm/m² of tank shell area.

Cooling water required for tank falling within same dyke area

$$\begin{aligned}\text{Cooling water rate} &= 3 \text{ lpm/m}^2 \text{ of tank area.} \\ \text{Cooling water required} &= \pi \times 40 \text{ m} \times 14.4 \text{ m} \times 3 \text{ lpm/m}^2 \\ &= 5431 \text{ lpm.} \\ &= \frac{5430.8 \times 60}{1000} \text{ m}^3/\text{hr} \\ &= 326 \text{ m}^3/\text{hr.}\end{aligned}$$

$$\begin{aligned}\text{Total fire water requirement for cooling of tanks (item i + ii)} &= 326 + 326 \\ &= 652 \text{ m}^3/\text{hr}\end{aligned}$$

b) Foam water flow rate

Water flow required for applying foam on a largest tank burning surface area (rim seal area)

For floating roof tank of 40 M diameter,

Diameter of the tank (D1) = 40M

Distance of foam dam from shell = 0.8M



$$\begin{aligned}\text{Diameter of roof up to foam dam (D2)} &= 40 - (2 \times 0.8) = 38.4 \\ \text{Rim seal area} &= (\pi / 4) \times (40^2 - 38.4^2) \\ &= (\pi / 4) \times 125.44 \\ &= 98.6 \text{ m}^2 \\ \text{Foam solution rate @ 12 lpm/ m}^2 &= 1183.2 \text{ lpm} \\ \text{Foam water required} &= 0.97 \times 1183.2 \text{ lpm} \\ \text{(For 3\% foam concentrate)} &= 1147.7 \text{ lpm.} \\ &= \frac{1147.7 \times 60}{1000} \text{ m}^3/\text{hr} \\ &= 69 \text{ m}^3/\text{hr.}\end{aligned}$$

Total water flow rate (item a + b) for floating roof tank protection

$$\begin{aligned}\text{Tank cooling} &= 652 \text{ m}^3/\text{hr.} \\ \text{Foam solution application} &= 69 \text{ m}^3/\text{hr.} \\ \text{Total water requirement} &= 721 \text{ m}^3/\text{hr.}\end{aligned}$$

3.2 FIRE WATER FLOW RATE FOR CONE ROOF TANK PROTECTION**Data**

$$\begin{aligned}\text{Total storage capacity in one dyke area} &= 50,000 \text{ m}^3. \\ \text{No. of tanks} &= 4. \\ \text{Capacity of each tank} &= 12,500 \text{ m}^3. \\ \text{Diameter of each tank} &= 37.5 \text{ m.} \\ \text{Height of each tank} &= 12 \text{ m.}\end{aligned}$$

a) Cooling water flow rate**(i) Cooling water required for tank on fire**

$$\begin{aligned}\text{Cooling water rate} &= 3 \text{ lpm/m}^2 \text{ of tank area for tank on fire.} \\ \text{Cooling water required} &= \pi \times 37.5 \text{ m} \times 12 \text{ m} \times 3 \text{ lpm/m}^2. \\ &= 4242.8 \text{ lpm.} \\ &= \frac{4242.8 \times 60}{1000} \text{ m}^3/\text{hr} = 255 \text{ m}^3/\text{hr.}\end{aligned}$$

Assuming that other three tanks are also located within the same tank dyke at a distance less than 30 m from the tanks shell. Therefore, in such case cooling required is at the rate of 3 lpm/m² of tank shell area.

(ii) Cooling water required for tanks falling within same dyke area

$$\begin{aligned}\text{Cooling water rate} &= 3 \text{ lpm/m}^2 \text{ of tank area.} \\ \text{Cooling water required} &= \pi \times 37.5 \text{ m} \times 12 \text{ m} \times 3 \text{ lpm/m}^2 \times 3 \\ &= 12728.6 \text{ lpm.} \\ &= \frac{12728.6 \times 60}{1000} \text{ m}^3/\text{hr} = 764 \text{ m}^3/\text{hr.}\end{aligned}$$

$$\begin{aligned}\text{Total cooling water required (item i + ii) for cone roof tank protection:} &= (255 + 764) \text{ m}^3/\text{hr} \\ &= 1019 \text{ m}^3/\text{hr}\end{aligned}$$

**b) Foam water flow rate**

Foam solution application rate	=	5 lpm/m ² of liquid surface area.
Foam solution required	=	$\pi \times (18.75 \text{ m})^2 \times 5 \text{ lpm/m}^2$.
	=	5524.5 lpm.
Foam water required	=	$0.97 \times 5524.5 \text{ lpm} = 5358.7 \text{ lpm}$.
(For 3% foam concentrate)	=	$\frac{5358.7 \times 60}{1000} \text{ m}^3/\text{hr}$
Total Foam water required	=	322 m³/hr.

Total water flow rate (item a + b) for cone roof tank protection

Tank cooling	=	1019 m ³ /hr.
Foam solution application	=	322 m ³ /hr.
Total	=	1341 m³/hr.

3.3 FIRE WATER FLOW RATE FOR COOLING POL TANK WAGON LOADING GANTRY**a) Data**

Total No. of loading points	=	Conventional or BTPN.
Width of tank wagon gantry (Cooling two spur)	=	12 m.

b) Cooling water flow rate

Divide total area of gantry into equal segments such that each segment measuring 15 m X 12 m and consider 3 segments operating at a time.

Water rate required	=	$3 \times 15 \text{ m} \times 12 \text{ m} \times 10.2 \text{ lpm/m}^2$.
	=	5508
	=	330 m³/hr

3.4 Fire water calculation for full surface fire on largest floating roof tank (roof sinking case)**Data:**

Total storage capacity in one dyke area	=	32,000 m ³
No. of tanks	=	2
Capacity of each tank	=	16,000 m ³
Diameter of each tank	=	40 m
Height of each tank	=	14.4 m

a) Cooling water requirement:



$$\begin{aligned}\text{Cooling water rate @ } 3 \text{ lpm/ m}^2 \text{ of tank shell area for tank-on-} \\ \text{fire Cooling water required} &= \pi \times 40 \times 14.4 \times 3 \\ &= 5430.9 \text{ lpm} \\ &= 326 \text{ m}^3/\text{hr}\end{aligned}$$

Assuming that second tank is located within the tank dyke at a distance more than 30M from the tank shell.

$$\begin{aligned}\text{Then, cooling water requirement @ } 3 \text{ lpm/ m}^2 \text{ of tank shell area} &= \pi \times 40 \times 14.4 \times 3 \\ &= 5430.9 \text{ lpm} \\ &= 326 \text{ m}^3/\text{hr}.\end{aligned}$$

$$\begin{aligned}\text{Total cooling water} &= (326 + 326) \text{ m}^3/\text{hr} \\ &= 652 \text{ m}^3/\text{hr}\end{aligned}$$

b) Water requirement in foam application

Foam Application Rate @ 9.75 lpm/m²

$$\begin{aligned}\text{Foam Solution Requirement} &= (\pi \times 40 \text{ m} \times 40 \text{ m}) / 4 \times 9.75 \text{ lpm/m}^2 \\ &= 12257.1 \text{ lpm} \\ &= 735.4 \text{ m}^3/\text{hr}\end{aligned}$$

$$\begin{aligned}\text{Water required for the foam solution} &= 0.97 \times 735.4 \text{ m}^3/\text{hr} \\ &= 713 \text{ m}^3/\text{hr}\end{aligned}$$

Total water required for roof sink case:

$$\begin{aligned}\text{Tank cooling} &652 \text{ m}^3/\text{hr} \\ \text{Foam application} &713 \text{ m}^3/\text{hr} \\ \text{Total} &1365 \text{ m}^3/\text{hr} \\ \text{Total water requirement} &= 1365 \text{ m}^3/\text{hr}\end{aligned}$$

Note: Full surface fire of floating roof tank roof sinking case being a remote possibility, it is considered as a single largest contingency for the purpose of arriving at design fire water requirement.

3.5 TOTAL DESIGN FIRE WATER FLOW RATE FOR TWO SIMULTANEOUS FIRE SCENARIO

The total fire water flow requirement will be sum of the two largest fire water requirement calculated in 3.1 (721 m³/hr) & 3.2 (1341 m³/hr), 3.3 (330 m³/hr) above i.e. (1341+721) = 2062 m³/hr plus supplementary water (36 x 4 = 144 m³/hr) = 2206 m³/hr.

OR water requirement alone as calculated in 3.4 i.e. 1365 m³/hr plus supplementary water (36 x 4 = 144 m³/hr) = 1509 m³/hr

Hence water requirement for double contingency locations shall be 2206 m³/hr

Note:- The fire water demand shall be determined considering 65 minutes (additional 30 minutes if foam stored for extended period) of foam application and 4 hours of tank cooling (including tank on fire) in the same dyke enclosure with supplementary water requirements for four hours. For storage capacity more than 100000 KL fire water for

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supplementary hose stream based on 4 hydrant streams + 1 High-Volume Long-Range water monitor. (i.e. 4 x 36 m³/hr + 1 x 228 m³/hr = 372 m³/hr) shall be considered refer clause 4.2.2(7).

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ANNEXURE - II

FIRE WATER FLOW RATE FOR COOLING POL TANK WAGON LOADING GANTRY

a) Data

Total No. of loading points = Conventional or BTPN.
 Width of tank wagon gantry = 12 m.
 (Cooling two spur)

b) Cooling water flow rate

Divide total area of gantry into equal segments such that each segment measuring 15 m X 12 m and consider 3 segments operating at a time

Water rate required = $3 \times 15 \text{ m} \times 12 \text{ m} \times 10.2 \text{ lpm/m}^2$.
 = 5508 lpm
 = **330 m³/hr**

a) **Water Requirement for supplementary Hose:**

Water for 4 single hydrant streams = $4 \times 36 = 144 \text{ m}^3/\text{hr}$.
Total water requirement = 144 m³ / hr.

Total water flow rate for gantry protection

(a) Gantry cooling = 330 m³/hr.
 (b) Supplementary hose requirement = 144 m³/ hr
Total = 474 m³/hr.

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ANNEXURE - III

SAMPLE CALCULATION OF FOAM COMPOUND REQUIREMENT FOR A DEPOT/ TERMINAL

1.0 FOAM COMPOUND CALCULATION FOR SINGLE FIRE CONTIGENCY

1.1 Foam compound calculation for single largest floating roof tank in a dyke.

Tank Data

Total storage capacity in one dyke area	=	20,000 m ³ .
No. of tanks	=	2.
Capacity of each tank	=	10,000 m ³ .
Diameter of each tank	=	30 m.
Height of each tank	=	14.4 m.

Foam compound requirement for tank

Foam solution application rate	=	12 lpm/m ² of rim seal area of tank.
Foam dam height	=	800 mm.
Diameter of the tank (D1)	=	30M
Distance of foam dam from shell	=	0.8M
Diameter of roof up to foam dam (D2)	=	30- (2X0.8) = 28.4
Rim seal area	=	$(\pi /4) \times (30^2 - 28.4^2)$
	=	$(\pi /4) \times 250.2$
	=	73.4 m ²
Foam solution rate @ 12 lpm/ m ²	=	880.8 lpm
3% Foam Compound required	=	26.4 lpm
Foam Compound required for 65 mins.	=	(26.4 x 65) litres = 1,716 litres

1.2 Foam compound calculation for single largest cone roof tank in a dyke.

Tank Data

Total storage capacity in one dyke area	=	10,000 m ³ .
No. of tanks	=	2
Capacity of each tank	=	5,000 m ³ .
Diameter of each tank	=	24 m.
Height of each tank	=	12 m.

Foam compound requirement for tank

Foam solution application rate	=	5 lpm/m ² of liquid surface area of tank.
Foam solution required	=	$(\pi \times (12)^2 \times 5)$ lpm = 2262.9 lpm
Foam compound required (3%)	=	0.03 x 2262.9 lpm = 67.9 lpm.
Foam compound required for 65 minutes	=	65 minutes x 67.9 lpm = 4,414 litres.

**1.3 FOAM COMPOUND CALCULATION FOR FLOATING TANK ROOF SINKING SCENARIO**

Capacity of Tank = 10,000 m³
Diameter of each tank = 30 m.
Height of each tank = 14.4 m.
Foam solution requirement = $((\pi/4) \times (30^2) \times 9.75)$ lpm
= 6894.6 lpm (1500 GPM
approx.) Nos. of HVLRs (assuming 750 GPM) = 750 GPM X 2 Nos.
Foam compound required (3%) = 0.03 x 6894.6 lpm = 206.83 lpm.
Foam compound required for 65 minutes = 65 minutes x 206.83 lpm = **13444 litres.**

1.4 FOAM COMPOUND CALCULATION FOR TWO HOSE STREAMS OF FOAM EACH WITH A CAPACITY OF 900 LPM.**Foam compound requirement for two foam hose streams of 900 lpm capacity**

Foam solution required = 2 x 900 lpm.
Foam compound required (3%) = 0.03 x 1800 lpm = 54 lpm.
Foam compound required for 65 minutes = 65 minutes x 54 lpm = **3510 litres.**

1.5 AGGREGATE QUANTITY OF FOAM COMPOUND FOR SINGLE FIRE CONTINGENCY

The aggregate quantity of foam solution shall be largest of the foam requirements calculated in 1.1 (1,716 litres), 1.2 (4,414 litres), 1.3 (13444 litres), 1.4 (3510 litres) i.e., 13444 litres

Therefore, foam compound to be stored = **13,444 litres**

2.0 FOAM COMPOUND CALCULATION FOR TWO MAJOR FIRES SIMULTANEOUSLY (For locations with aggregate storage capacity more than 30000KL)**2.1 Foam compound calculation for single largest floating roof tank in a dyke.****Tank Data**

Total storage capacity in one dyke area = 32,000 m³.
No. of tanks = 2.
Capacity of each tank = 16,000 m³.
Diameter of each tank = 40 m.
Height of each tank = 14.4 m.

Foam compound requirement for tank

Foam solution application rate = 12 lpm/m² of rim seal area of tank.
Foam dam height = 800 mm.
Diameter of the tank (D1) = 40 m
Distance of foam dam from shell = 0.8 m
Diameter of roof up to foam dam (D2) = 40 - (2X0.8) = 38.4 m
Rim seal area = $(\pi/4) \times (40^2 - 38.4^2)$
= 98.6 m²



Foam solution rate @ 12 lpm/ m² = 1183.2 lpm
3% Foam Compound required = 35.5 lpm
Foam Compound required for 65 mins. = **2308 litres**

2.2 Foam compound calculation for single largest cone roof tank in a dyke.

Tank Data

Total storage capacity in one dyke area = 50,000 m³
No. of tanks = 4.
Capacity of each tank = 12,500 m³.
Diameter of each tank = 37.5 m.
Height of each tank = 12 m.

Foam compound requirement for tank

Foam solution application rate = 5 lpm/m² of liquid surface area of tank.
Foam solution required = $(\pi \times (18.75)^2 \times 5)$ lpm = 5524.6
Foam compound required (3%) = 0.03 x 5524.6 lpm = 165.7 lpm.
Foam compound required for 65 minutes = 65 minutes x 165.7 lpm = **10,771 litres.**

2.3 In case of double fire, foam will be required to pour in both tanks simultaneously.
Foam compound requirement for 65 minutes = (2308 + 10,771) litres = **13,079 litres.**

2.4 FOAM COMPOUND CALCULATION FOR ROOF SINKING SCENARIO

Foam application rate of 9.75 lpm/m²

Data:

Total storage capacity in one dyke area = 32,000 m³.
No. of tanks = 2.
Capacity of each tank = 16,000 m³.
Diameter of each tank = 40 m.
Height of each tank = 14.4 m.

Foam solution requirement = $((\pi / 4) \times (40^2) \times 9.75)$ lpm
= 12257 lpm (2690 GPM)
Nos. of HVLRs (assuming 1000 GPM) = 1000 GPM X 3 Nos.

Foam compound required (3%) = (12257 x 0.03) lpm
= 367.7 lpm
Foam compound required for 65 minutes = 65 minutes x 367.7 lpm = **23900 litres.**
Say 24000 litres

2.5 FOAM COMPOUND CALCULATION FOR TWO HOSE STREAMS OF FOAM EACH WITH A CAPACITY OF 1140 900 LPM.

Foam compound requirement for two foam hose streams of 900 lpm capacity

Foam solution required = 2 x 900 lpm.
Foam compound required (3%) = 0.03 x 1800 lpm = 54 lpm.
Foam compound required for 65 minutes = 65 minutes x 54 lpm = **3510 litres.**

**2.6 AGGREGATE QUANTITY OF FOAM COMPOUND FOR TWO FIRES
SIMULTANEOUSLY**

The aggregate quantity of foam solution shall be largest of the foam requirements calculated in 2.3 (13,079 litres), 2.4 (24000 litres) or 2.5 (3510 litres) i.e. **24,000 litres**

Therefore, foam compound to be stored = 24,000 litres

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**ANNEXURE-IV****BRIEF DESCRIPTION OF FIRE FIGHTING FOAM****1.0 FIRE FIGHTING FOAM**

Firefighting foam is a homogeneous mass of tiny air or gas filled bubble of low specific gravity, which when applied in correct manner and in sufficient quantity, forms a compact fluid and stable blanket which is capable of floating on the surface of flammable liquids and preventing atmospheric air from reaching the liquid.

2.0 TYPES OF FOAM COMPOUND

Two Types of foams are used for fighting liquid fires:

2.1 CHEMICAL FOAM

When two or more chemicals are added the foam generates due to chemical reaction. The most common ingredients used for chemical foam are sodium bicarbonate and aluminium sulphate with stabilizer. The chemical foam is generally used in fire extinguishers.

2.2 MECHANICAL FOAM

It is produced by mechanically mixing a gas or air to a solution of foam compound (concentrate) in water. Various types of foam concentrates are used for generating foam, depending on the requirement and suitability. Each concentrate has its own advantage and limitations. The brief description of foam concentrates is given below.

3.0 TYPES OF MECHANICAL FOAM

Mechanical foam compound is classified into 3 categories based on its expansion ratio.

3.1 LOW EXPANSION FOAM

Foam expansion ratio can be up to 50 to 1, but usually between 5:1 to 15:1 as typically produced by self-aspirating foam branch pipes.

The low expansion foam contains more water and has better resistant to fire. It is suitable for hydrocarbon liquid fires and is widely used in oil refinery, oil platforms, petrochemical and other chemical industries.

3.2 MEDIUM EXPANSION FOAM

Foam expansion ratio vary from 51:1 to 500:1 as typically produced by self-aspirating foam branch pipes with nets. This foam has limited use in controlling hydrocarbon liquid fire because of its limitations w. r. t. poor cooling, poor resistant to hot surface/radiant heat, etc.



3.4 HIGH EXPANSION FOAM

Foam expansion ratio vary from 501:1 to 1500:1, usually between 750:1 to 1000:1 as typically produced by foam generators with air fans. This foam also has very limited use in controlling hydrocarbon liquid fire because of its limitations w. r. t. poor cooling, poor resistant to hot surface/radiant heat, etc. It is used for protection of hydrocarbon gases stored under cryogenic conditions and for warehouse protection.

4.0 TYPES OF LOW EXPANSION FOAM

4.1 PROTEIN FOAM

The foam concentrate is prepared from hydrolysed protein either from animal or vegetable source. The suitable stabilizer and preservatives are also added.

The concentrate forms a thick foam blanket and is suitable for hydrocarbon liquid fires, but not on water miscible liquids. The effectiveness of foam is not very good on deep pools or low flash point fuels which have had lengthy preburn time unless applied very gently to the surface.

The concentrate is available for induction rate of 3 to 6%. The shelf life of concentrate is 2 years.

4.2 FLUORO PROTEIN FOAM

This is similar to protein base foam with fluoro-chemical which makes it more effective than protein base foam.

The concentrate forms a thick foam blanket and is suitable for hydrocarbon liquid fires, but not on water miscible liquids. The foam is very effective on deep pools of low flash point fuels which have had lengthy pre burn time.

The concentrate is available for induction rate of 3 to 6% and the shelf life is similar to that of protein base foam.

4.3 AQUEOUS FILM FORMING FOAM (AFFF)

The foam concentrate mainly consists of fluoro carbon surfactants, foaming agent and stabilizer. This can be used with fresh water as well as with sea water.

It produces very fluid foam, which flows freely on liquid surface. The aqueous film produced suppresses the liquid vapor quickly. The foam has quick fire knock down property and is suitable for liquid hydrocarbon fires. As the foam has poor drainage rate, the effectiveness is limited on deep pool fires of low flash point fuels which have lengthy pre burn time.



The concentrate is available for induction rate of 3 to 6% and the shelf life is more than 10 years. This can also be used with non-aspirating type nozzles.

4.4 MULTIPURPOSE AFFF

Multipurpose AFFF concentrate is synthetic, foaming liquid designed especially for fire protection of water soluble solvents and water insoluble hydrocarbon liquids. This can be used either with fresh water or sea water.

When applied it forms foam with a cohesive polymeric layer on liquid surface, which suppresses the vapor and extinguishes the fire. The foam is also suitable for deep pool fires because of superior drainage rate and more resistive to hot fuels/radiant heat.

The 3% induction rate is suitable for liquid hydrocarbon fires and 5% for water miscible solvents. The shelf life of concentrate is not less than 10 years. This can also be used with non-aspirating type nozzles.

4.5 FILM FORMING FLOURO PROTEIN FOAM (FFFPF)

FFFPF combines the rapid fire knock down quality of conventional film forming AFFF with the high level of post fire security and burn back resistance of flouro-protein foam. The concentrate can either be used with fresh water or sea water.

The foam is suitable for hydrocarbon liquid fires including deep pool fires of low flash point fuels which have had lengthy pre burn time.

The concentrate is available for induction rate of 3 to 6% and the shelf life is 5 years. This can also be used with non-aspirating type nozzles.

5.0 TYPES OF MEDIUM AND HIGH EXPANSION FOAM

Synthetic foam concentrate is used with suitable devices to produce medium and high expansion foams. This can be used on hydrocarbon fuels with low boiling point. The foam is very light in weight and gives poor cooling effect in comparison to low expansion foams. The foam is susceptible to easy break down by hot fuel layers and radiant heat.

The induction rate in water should vary from 1.5 to 3%. Many of the low expansion foam concentrate can also be used with suitable devices to produce medium / high expansion foam.

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ANNEXURE- V

SYSTEM OF AUTOMATIC ACTUATED RIM SEAL FIRE DETECTION AND EXTINGUISHING SYSTEM FOR EXTERNAL FLOATING ROOF TANKS STORING CLASS- A PETROLEUM

The automatic actuated foam flooding system is a system designed to automatically detect and extinguish the floating roof tank rim seal fire at its incipient stage. The system is mounted on the roof of the tank. The minimum requirement for the design of the system is given below:

1.0 FOAM FLOODING SYSTEM

Film Forming Fluro Protein Foam (FFFP) / Aqueous Film Forming Foam (AFFF)/ Alcohol resistant Aqueous Film Forming Foam (AR-AFFF) type concentrate is used in the system.

1.1 FOAM APPLICATION SYSTEM

A large storage tank require one or more than one modular units for foam application in the entire rim seal. Each such unit consists of a foam distribution pipe, laid along the tank perimeter over the rim seal area. The spray nozzles for foam application are mounted on the distribution pipe at suitable intervals. Distribution pipe is permanently connected to a storage vessel containing pre-mix foam and both are placed on the roof. The premix foam is kept pressurized with nitrogen/ CO₂ in a vessel. The system is designed for minimum foam application rate of @ 18 lpm/ m² of rim seal area. For effective control, foam is discharged in approximately 40 seconds.

1.2 ALARM & AUTO ACTUATION SYSTEM

In case of fire on the rim seal, it is automatically detected by a device capable to sense the same. The device then actuates the spray system for application of foam in the complete area of rim seal to quickly extinguish the fire in its incipient stage. An audio-visual alarm is also coupled with the detection & extinguishing system for necessary fire alert.

The system includes a fire detection system which senses fire and actuates the automatic release of the extinguishing medium on the rim seal area. Each tank shall have independent detection & extinguishing system.

The validity of the approach must be demonstrated by the designer for an effective total flooding extinguishing system which quickly detects and extinguishes fire in its incipient stage without re-flash. Also, the design considerations should include the impact of the weight of the modules placed on the floating roof.

The detection system needs to be highly reliable and shall work at varied site ambient temperatures for protection of rim seal fire.

These detection systems shall be listed and/ or approved by any of the national or



international agencies like BIS, UL, FM, VdS or LPC to ensure that those systems are used which meet with highest standards of safety.

1.3 CALCULATIONS FOR MODULAR FOAM APPLICATION SYSTEM

Rim seal area of tank = $\pi \times 79 \text{ m} \times 0.3$

$\text{m} = 74.5 \text{ m}^2$ (Considering a flexible seal width of typically 300 mm)

Foam solution application rate 18 lpm/m^2 =

1341

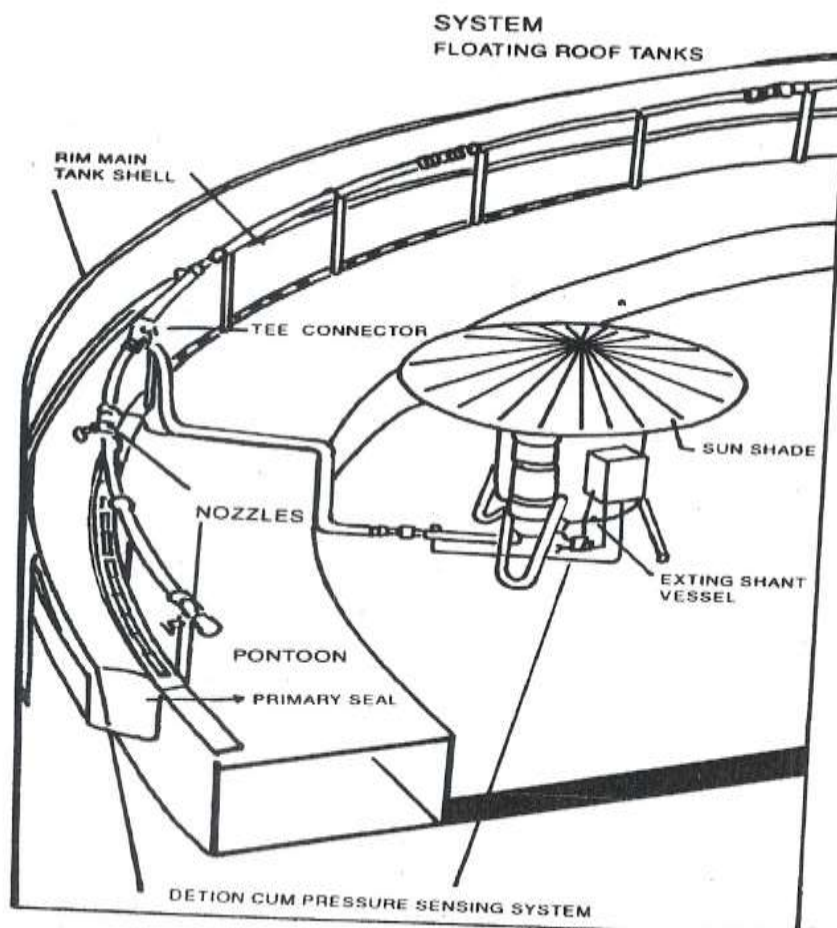
Liters. Total foam solution required in 40 seconds =

894

Liters. Total nos. of modular unit required =

7 Nos.

(Considering a vessel of 150 Liters capacity containing 135 Liters of foam)



**Annexure –VI****EMERGENCY KIT**

Emergency Kit consists of listed emergency equipment required for rescue and control/arresting leakage in case of emergency in oil terminals & depots. The equipment shall be mounted on a compact light weight trolley. Emergency Kit shall be consisting of the following emergency equipment:

S.N o.	Item	Quantity	Remarks
1	COLD/LOW TEMPERATURE PROTECTIVE SUIT.	2 sets.	For LPG locations
2	FIRE PROXIMITY SUIT	1 set	
3	PVC SUIT	2 sets	
4	LEAK CONTROL KIT	Consisting of 1 no each of leak arresting pad, leakage control of external pipes, internal pipes, large external pipes up to 8 inch, drums / containers leakages, general purpose leakages, large hole leakages in storage tanks. - 1 set	
5	PETROLEUM PRODUCT CLEANUP CHEMICAL - Boom(5 inch dia , 3 mtr. Long) : 6 nos. - Boom(3 inch dia , 3 mtr. Long) : 6 nos. - Granular particles to absorb Oil : 20 Kg	1 set.	
6	OIL SPILL DISPERSANT (WATER BASED) along with handheld spray nozzle.	Dispersant : 40 litre Spray Gun with backpack : 1 set.	



**FIRE PROTECTION FACILITIES FOR PETROLEUM
DEPOTS, TERMINALS, CAVERNS, CTF PIPELINE
INSTALLATIONS AND LUBE OIL INSTALLATIONS**

7	NON SPARKING TOOLS One set consisting of : - Shoe handle brush -01 no - 9" Crate opener -01 no - 16oz Claw hammer with Fiberglass handle -01no -Common knife 5 ³ / ₄ " Blade : 1 no 10 ³ / ₄ " OAL, -12" Groove joint plier, -7" Long nose pliers with cutters, -8" Combination Pliers, -Deck scraper, -1 ¹ / ₂ " Blade X 15"Long, -Spray booth scraper, -3" blade X 9 ¹ / ₄ " Long, -Std Screwdriver – ⁵ / ₁₆ " Tip, -6" Blade, - 3" Phillips Screwdriver, -12" Tin Snips, -8" Adjustable Wrench, -12" Adjustable wrench, -14" pipe Wrench (Aluminium), -12" Bung Wrench (Fits ³ / ₄ " X 2")	One Set.	
8	BREATHING APPARTUS SET (40 minute duration)	2 sets	
9	EMERGENCY ESCAPE SETS (15 minutes duration)	2 Sets	
10	FLAME PROOF SEARCH LIGHT	2 no's	Rechargeable type suitable for Explosive Environment.
11	MEGAPHONE EX-PROOF	1 set	Portable battery operated PA System with 1 loudspeaker with a range of 1 KM in still air and 500 M in noisy areas.
12	HAND SIREN WITH STAND	1 no	Approx. range of 1.6 KMS
13	FIREMAN AXE	1no	
14	FIRST AID BOX	1 no.	
15	MANUAL RESCUCIATOR	1 no	Manually operated for artificial respirators

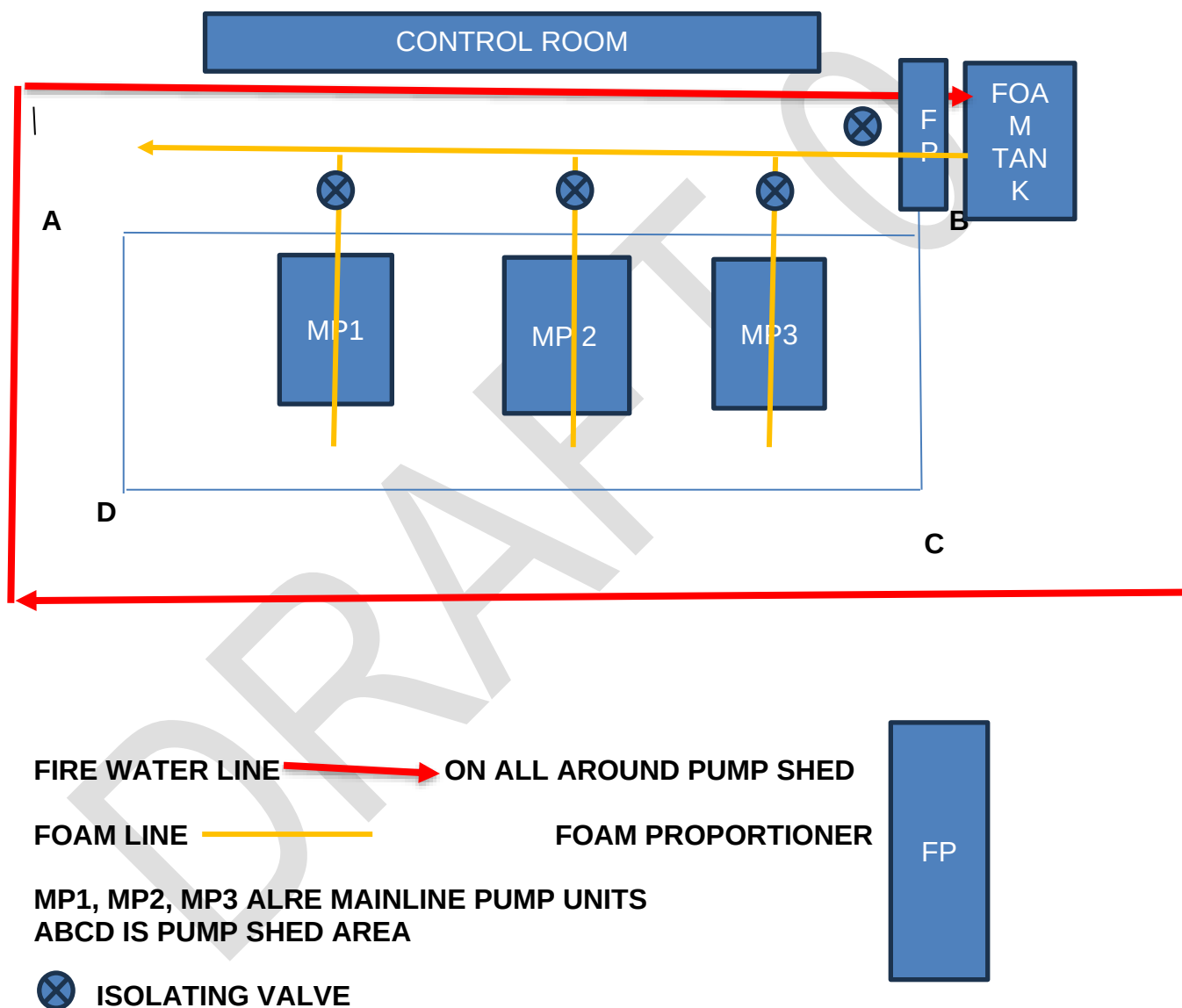


S.N o.	Item	Quantity	Remarks
			consisting of adult size nose, mouth, face plate, air bulb with oxygen inlet connection, non-return, non-breathing human valves and first aid charge packed in a plastic bag
16	FOLDING STRETCHER	1 no	Size 6 feet X 3 feet with tying belts & blanket.
17	MECHANICAL TOOL KIT	1 set	
18	COLD / LOW TEMPERATURE HAND GLOVES	4 Pairs	
19	ELECTRICAL RUBBER HAND GLOVES	2 pair	Suitable for Electrical jobs upto 33000 Volts
20	ELECTRICAL TESTER	1 no	
21	CHEMICAL/OIL SPLASH PROOF GOGGLES	4 nos.	ANSI/CE marked

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ANNEXURE-VII

Typical schematic diagram of firefighting facility for Pipeline Mainline pump shed

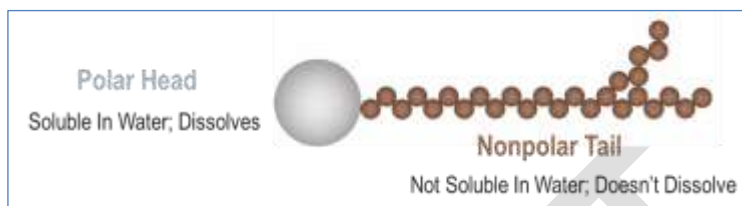


ANNEXURE-VIII

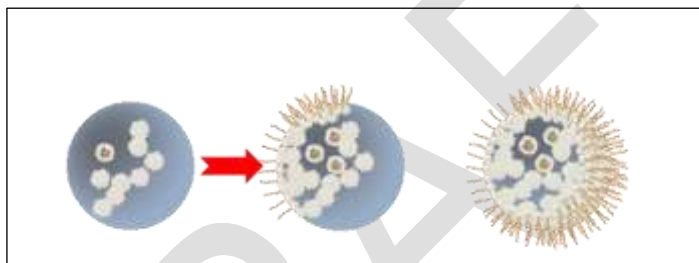
Encapsulator Agent

Encapsulator Agents are a type of water additive whose basic building blocks are spherical micelles. While most of the conventional firefighting media separate fuel from oxygen, Encapsulator Agents work on a molecular level. Encapsulator Agents alter the chemical makeup of plain water droplets, leading to several advantages.

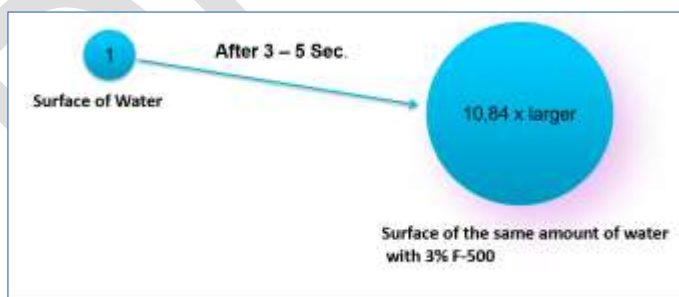
Encapsulator Agents start with a large amphipathic molecule, containing both a hydrophilic polar head and hydrophobic non-polar tail that act independently.



When mixed with water, these molecules will instantly form billions of micro spherical micelles as non-polar tails turn inward, avoiding water at all costs.



Encapsulator Agents immediately reduce the surface tension of water. This makes droplets smaller, creating more surface area to rapidly absorb heat and better penetrate the pores of solid material.



As Encapsulator Agent (EA) droplets leave the nozzle, non-polar tails near the surface turn outward, forming a protective “skin” around each droplet. This happens instantly, leaving a multitude of spherical micelles within each EA droplet, changing the EA droplet’s heat reduction mechanism.



This enables an EA droplet to absorb significantly more heat than a plain water droplet. Thermal energy is driven inward towards the countless spherical micelles inside. A plain water droplet lacks the ability to absorb heat three -dimensionally and therefore generates scalding steam. This feature allows Encapsulator Agents to extinguish fires more efficiently than plain water and gives them the ability to safely mitigate fires that plain water cannot. In addition to this, the non-polar tail of EA molecule completely wraps around / encapsulates any organic molecule present in the fuel and by-products of fire like smoke and off-gases. By encapsulating the fuel, it makes it chemically unavailable for further reduction and thus extinguishing fire.

-----) 0 x 0 (-----

**ANNEXURE-IX****MAJOR CHANGES IN THE REVISED EDITION**

- 2.1** This standard lays down the minimum requirement of fire protection facilities at Petroleum Depots, Terminals, Caverns, Pipeline Installations with or without Storages, Cross country natural gas pipeline stations, sectionalizing valve stations, Compressor stations, Central Tank Farms (CTF), Lube Oil Installations, Grease Manufacturing & Filling Facilities. The fire protection and firefighting requirement for the combined POL and LPG facilities in the same premises.
- 2.2** This standard shall override the requirements of fire protection facilities specified in any other OISD standard for the installations mentioned in Clause 2.1.
- 4.2.2** Fire protection system shall be provided based on aggregate storage capacity as per below mentioned table:
- 4.2.4 Fire protection system based on aggregate storage capacity:**
- 4.2.6 Fixed Water Spray on storage Tanks**
- 4.2.7** Semi-fixed foam system or Fixed foam system shall be provided for the following tanks:
- Floating roof tanks storing Class 'A' and Class 'B' petroleum products.
 - Fixed roof tanks storing Class 'A' and class 'B' petroleum products.
 - Fixed roof tanks storing class 'C' petroleum products, of diameter larger than 40 M.

Foam System should not be provided for fixed roof tanks operating at more than 0.6 psi pressure and where the tanks are nitrogen or other inert gas blanketed. In fixed roof tanks storing high-viscosity liquids in temperature above 93.3°C (200°F), foam system should not be provided.

4.3.1.1 DESIGN FLOW RATE**4.3.3 STORAGE****4.3.5 FIRE WATER PUMPS****4.4.3.3 Protection using Automatic fire detection and Foam suppression System:**

Automatic actuated Rim Seal fire detection and extinguishing system shall be provided on all external floating roof tank storing Class A Petroleum products. The components of the system shall be listed or approved by national or international agencies such as BIS, UL and FM etc. The minimum requirement of design for the automatic actuated foam flooding system is given in Annexure V.

4.4.3.4 Protection using Fixed / Semi-Fixed Foam System

- I. Fixed Foam suppression system should be fully automatic comprising foam tank, inline induction, foam proportioning and fixed piping to be operated on automatic detection of fire at rim seal area.
- II. In case of Semi-Fixed foam system, foam inlet breeching connections shall be located at accessible location.



- III. System shall be designed to create foam blanket on the burning surface in a reasonably short period.
- IV. Continuous foam concentrate supply shall be applied to the burning hazard at a rate high enough to overcome the destructive effects of radiant heat by Centralized foam system / multiple foam nurseries.
- V. Foam makers/foam pourers shall be located not more than 24 M apart on the shell perimeter based on 600 mm foam dam height. The height of foam dam shall be at least 51 mm above the top of metallic secondary seal.
- VI. A minimum of two foam pourers shall be provided.
- VII. Whenever mobile type of systems like foam nurseries are used for supplying foam to HVLR of storage tank, provision shall be made to connect more than one foam nurseries for continuous foam concentrate supply.

4.4.4 FIXED ROOF TANK PROTECTION

- iv) Tanks should be provided with foam discharge outlets/pourers as indicated below :-

Tank Diameter in m.	Minimum number of foam discharge outlet
> 9 upto 20	2

4.4.6 FULL SURFACE TANK FIREFIGHTING & LOGISTICS

4.4.7 FOAM APPLICATION RATE

In the case of Floating roof tank roof sinking, the application rate shall be considered as 9.75 lpm/ m².

4.4.8 DURATION OF FOAM DISCHARGE

Tanks containing liquid hydrocarbons - Class 'C' Petroleum - 30 minutes.

Note: - It is advised to keep foam compound stock for extended 30 minutes for full surface tank fire scenario.

4.4.11 FOAM QUANTITY REQUIREMENT

4.4.12 FOAM COMPOUND STORAGE

4.5 FIREFIGHTING ON FIRES RELATED TO ETHANOL

4.7.2 Wheeled Fire Fighting Equipment

6.0 SAFETY, FIRE PROTECTION SYSTEM AND EMERGENCY RESPONSE FOR CROSS COUNTRY PIPELINES

7.0 BATTERY FIRE

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Shri Shashi Dua	Indian Oil Corporation Limited
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Shri D. K. Banerjee	Indian Oil Corporation Limited
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